

The effect of layer thickness and fluid conductivity contrasts on seismoelectric signals: insights from laboratory experiments and numerical simulations

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SUMMARY

Seismoelectromagnetic (SE) signals are created in wet poroelastic media by electrokinetic conversions occurring at the pore scale. Two of these signals are frequently investigated: a co-seismic wavefield, bounded to the propagating seismic waves and an electromagnetic (EM) wave created when a seismic wave passes through the interface between two porous media. Driven by the possibility offered by the SE signals in terms of vertical resolution, compared with seismic reflection methods, a restricted number of authors studied how a thin layer, or a combination of thin layers, can modify SE signals. In this context, this work presents original experimental and numerical data as a means to further investigate the relation between the thickness of a layer and the EM interface-generated response (IR). In addition, the fact that the IR is very sensitive to contrasts in fluid conductivity, whereas seismic waves are not, is an appealing characteristic of SE exploration. Consequently, we explored the influence of the pore-fluid electric conductivity, combining experimental and numerical methodologies. We identified a strong similarity in how layer thinning affects both seismic waves and IR signals. Moreover, we observed that the maximum enhancement of an IR signal occurs when the layer thickness is approximately half the *P*-wave wavelength (λ_P) in the layer. When analysing the influence of fluid conductivity, we showed that the electrokinetic theory adopted in this study provides satisfactory predictions for the waveforms and amplitudes observed experimentally. Finally, we extrapolated our experimental results with field-scale simulations in order to understand how the effects observed experimentally translate to georesources applications.

Key words: Electrical properties; Electromagnetic theory; Numerical modelling; Numerical solutions; Hydrogeophysics.

1 INTRODUCTION

During the propagation of mechanical waves in porous media containing pore fluids, seismic-to-electromagnetic energy conversion, conventionally termed seismoelectric phenomena, takes place due to the differential movement of mobile ions in the fluid relative to counterions adsorbed at the solid grain boundaries, which is caused by the relative fluid–solid displacement that accompanies the passage of a seismic disturbance. Additionally, the inverse electromagnetic-to-seismic energy conversion, obeying the same electrokinetic physics, will also exist when EM waves disturb a porous medium. R.R. Thompson (1936) was the first to formally draw attention to the potential

for developing a geophysical method to study such phenomena. Since then, these electrokinetic conversions became the focus of various field studies (e.g. S.T. Martner & N.R. Sparks 1959; R.D. Russell *et al.* 1997; S. Garambois & M. Dietrich 2001; J.C. Dupuis *et al.* 2009; W. Rabbel *et al.* 2020), laboratory experiments (e.g. Z. Zhu *et al.* 1999; C. Bordes *et al.* 2006; M.D. Schakel *et al.* 2011; J. Wang *et al.* 2020; J. Liu *et al.* 2024; C. Nie *et al.* 2025) and numerical investigations (e.g. S. Garambois & M. Dietrich 2002; F.I. Zyserman *et al.* 2010; H. Ren *et al.* 2010; N. Grobbo *et al.* 2016; D. Wang *et al.* 2023; J. Li *et al.* 2024; L. Han *et al.* 2024). Regarding the theoretical developments, S. Pride (1994) developed a set of equations capable of explaining and modelling the coupling between mechanical and electromagnetic wavefields

in fluid-saturated isotropic porous media. His theory remains, to this date, the most widely accepted. Nevertheless, recent theoretical formulations (e.g. A. Revil & H. Mahardika 2013; D. Jougnot *et al.* 2013; A. Jardani & A. Revil 2015; M. Rosas-Carbajal *et al.* 2020; D. Jougnot & S.G. Solazzi 2021; L.B. Monachesi *et al.* 2023; Y. Zhao *et al.* 2023) appear as alternative solutions, worthy of consideration, to describe seismoelectromagnetic phenomena for a wider class of porous media.

S. Pride (1994) adopted the thin electric double-layer theory (O. Stern 1924; J.O. Bockris & A.K.N. Reddy 1970) and an averaging theorem (J.C. Slattery 1967; S. Whitaker 1969) to combine the microscopic forms of Maxwell's and Biot's equations to describe the aforementioned physics at the macroscale. S.R. Pride & M.W. Haartsen (1996) complemented the latter by presenting solutions for point-sources and plane-waves in a homogeneous isotropic whole space, inferring the existence of four wave types: longitudinal seismic (slow and fast-P) waves, one transverse seismic wave (shear or S) and one transverse EM-field; they also derived the boundary conditions that hold at the interface of distinct porous media. Finally, M.W. Haartsen & S.R. Pride (1997) provided a numerical code able to simulate all coupled mechanical and electromagnetic wavefields according to Pride's theory.

Following these developments, three types of seismoelectromagnetic signals are usually studied: the coseismic fields, intrinsically linked to seismic waves (C. Bordes *et al.* 2008; J. Holzhauer *et al.* 2017; Y.C. Sun *et al.* 2019; X. Ma *et al.* 2022); the IRs, radiated from discontinuities inside the porous media or created directly at the interface between the source and the porous medium (K.E. Butler *et al.* 1996; Z. Zhu *et al.* 2000; J.C. Dupuis *et al.* 2007; M. Schakel & D. Smeulders 2010; D. Smeulders *et al.* 2014; D. Wang *et al.* 2021); and evanescent EM waves, that are generated whenever the seismic waves reach an interface with an incidence angle greater than the critical angle (H. Ren *et al.* 2016, 2018; K.E. Butler *et al.* 2018; L. Dzieran *et al.* 2019; S. Yuan *et al.* 2021; X.Z. Zheng *et al.* 2021; K. Hu *et al.* 2023). A fourth SE signal has recently gained attention, partly due to its potential for geothermal exploration (C. Morency *et al.* 2021), namely, the IR created at elastic-poroelastic interfaces (F. Bucher *et al.* 2023; N. Bernardo *et al.* 2024; Y. Kang *et al.* 2024). Here, we discuss the first two types, but focusing on the EM interface response, and with special attention to how a thin layer (layers with thickness smaller than λ_p in the medium) and contrasts of fluid conductivity may influence these signals.

For a long time, thin layers have drawn the attention of environmental and exploration geophysicists, for seismic and ground-penetrating radar applications, regarding their detectability, amplitude variation with offset and an amplitude boosting effect (M.B. Widess 1973; R. Tatalovic *et al.* 1988; Y. Liu & D.R. Schmitt 2003; J.H. Bradford & J.C. Deeds 2006; J. Deparis & S. Garambois 2009; A. Shakas & N. Linde 2015; G. Papageorgiou & M. Chapman 2021). The effect of thin layers on the EM interface response has mainly focused on evaluating its amplitude-boosting potential, in order to overcome the low signal-to-noise ratio often associated with this SE signal (S. Garambois & M. Dietrich 2001). M.W. Haartsen & S.R. Pride (1997) first discussed the amplitude boosting of seismoelectric waves created by replacing a thick (relative to λ_p) sand layer with a thin one. S.R. Pride & Garambois (2005) arrived at the same conclusion about the potential of thin layers to enhance

seismoelectric interface-generated waves, even though the bulk increase detected in their study was not only an effect of the thickness of the layer, but was also due to the high permeability contrast created by the thin layer. S.S. Haines & S.R. Pride (2006) used a 2-D finite-difference modelling to study the influence of layer thickness and lateral extent on the EM interface response. They conclude that the seismoelectric method is effective to detect thin (regarding λ_p) and narrow (with respect to the first Fresnel zone radius) layers. N. Grobbee & E.C. Slob (2016) investigated the EM response of a composite layer consisting of a stack of thin-beds, with a focus on the S-to-EM wave conversion created by the layering. For that, the thickness of the stratification and of the individual beds inside were varied. Regarding the effect of the total layer thickness, the authors found that the highest amplitudes occurred when the thickness was close to either the P- or the S-wave wavelengths. To our knowledge, the only experimental work addressing the effect, on the EM interface response, of a thin layer embedded in a homogeneous medium is that of R. Peng *et al.* (2017). Their experiment employed a wedge-shaped rock sample and their results showed that the amplitude of the EM interface response increased with thinning of the wedge. More recently, Y. Kang *et al.* (2024) numerically studied the effect of thin porous layers sandwiched between solid half-spaces.

The electric conductivity of the fluids filling up the pores of the geological layers forming Earth's crust is often studied by geophysicists to infer the existence of geothermal reservoirs (J. Domra Kana *et al.* 2015), assess groundwater resources (M. Goldman & F. Neubauer 1994; H. Nouradine *et al.* 2024), detect the presence of contaminants in the soil (S. Garambois *et al.* 2002; P. Sénéchal *et al.* 2005) or identify and monitor fluid types in oil-bearing reservoirs (J.H. Saunders *et al.* 2008; J. Guo *et al.* 2023). In this context, the SE method shows a great potential for geophysical studies, since the electric conductivity of the pore-fluid has a strong influence on the amplitudes of the measured signals. More specifically, the EM interface response can be used to detect fluid conductivity contrasts with the resolution of seismic methods whereas seismic waves are insensitive to these changes in the subsurface (see M.W. Haartsen & S.R. Pride 1997).

In order to investigate the dependence of SE signals on the fluid conductivity (σ_f), Z. Zhu & M.N. Toksöz (2003) used a borehole model immersed in a water tank with a fracture separating two different rocks. By varying the fluid conductivity and analysing the IRs created at the two interfaces of the fracture, they identified two pore-fluid electric conductivity regions, based on how the IR amplitude varied with increasing σ_f . The first one corresponds to a σ_f range where the IR amplitudes increase with increasing σ_f . This is explained by a Stern layer in the material's pores (S.R. Pride & F.D. Morgan 1991; A. Revil & N. Florsch 2010) that does not contain enough ions adsorbed to the solid-grain surface, that is, that is incomplete. Thus in this σ_f range, increasing the number of ions available will favour SE coupling. The second one refers to a higher σ_f range where IR amplitudes decrease with increasing conductivities. Such a decrease is due to conduction currents, whose magnitude is a function of σ_f , becoming dominant over the streaming electric currents in the pores. As a consequence, the amplitudes of the SE signals will drop proportionally to the molarity of the fluid (S. Pride 1994). B. Chen & Y. Mu (2005), G.I. Block & J.G. Harris (2006), M.D. Schakel *et al.* (2011), Z. Zhu & M. Nafi Toksöz

(2013) and J. Holzhauer *et al.* (2017) have also observed a decrease of SE signals as the fluid conductivity increases using different experimental configurations, porous media and range of fluid conductivities. These studies agree that the electrokinetic theory of S. Pride (1994) is capable of predicting their experimental measurements satisfactorily. However, when comparing experimental data and numerical simulations, G.I. Block & J.G. Harris (2006) and M.D. Schakel *et al.* (2011) noted that by decreasing the fluid conductivity, the discrepancy between experimental data and numerical simulations based on Pride's equations increases.

This work presents new experimental and numerical data which allowed us to further investigate the influence of layer thinning on the EM interface response, and to study the effect of varying electrical conductivities on SE wavefields. We use a unique experimental setup to simultaneously study both seismic and SE signals, and perform numerical simulations following the work of V. Martins-Gomes *et al.* (2023). While most experimental studies on SE conversions consider a rock sample inside a water tank, our experiments involve two porous media in contact with each other, a configuration considered to be more representative of real-life situations encountered in the geophysical characterization of the subsurface. To begin with, we will concentrate on the following questions: (i) what is the effect of layer thickness on the EM IR? (ii) how does this effect, if it exists, modify the IR amplitudes? Then, we will assess the effect of fluid conductivity to complement previous studies carried out on the subject by providing a quantitative analysis of how co-seismic and IR signals depend on this fluid property. At last, we contextualize our experimental results using field-scale numerical simulations.

2 BACKGROUND CONCEPTS

2.1 On the seismic and EM responses of a thin slab

Considering seismic reflections generated at normal incidence at the two opposite interfaces of an infinitely long horizontal layer sandwiched between two identical half-spaces, M.B. Widess (1973) defined a thin bed as a layer whose thickness d is less than about one-eighth of the dominant wavelength λ_p inside the layer. He also emphasized that thin layers may have significantly higher reflectivity than is generally attributed to them. For very thin beds ($d < \lambda_p/8$), he noted that the individual top and bottom reflections merge into a single event whose composite signature is the time-derivative of the incident waveform. Generally speaking, when the thickness d of a layer embedded in a uniform medium is smaller than the dominant seismic wavelength, constructive and destructive interference taking place inside the layer influence the reflected and transmitted waves. As a consequence, both the seismic amplitudes and waveforms change with layer thinning.

These results can be described with the well-established stationary time-invariant convolutional model (E.A. Robinson 1985) in which the attenuation inside the media is neglected. Disregarding the energy partitioning at the interfaces, the normal incidence seismic response of a thin layer $s(t)$ of thickness d at a location in the upper half-space can be represented, in the time-domain (t), as a combination of two opposite-polarity and equal magnitude reflections as:

$$s(t) = w(t) * [\delta(t) - \delta(t - \tau)], \quad (1)$$

where $w(t)$ is the source wavelet function, $*$ is the convolution operator, $\delta(t)$ is the Dirac delta function, τ is the time delay between the contributions generated at the two boundaries of the layer.

In the frequency domain, eq. (1) becomes:

$$S(\omega) = W(\omega) [1 - e^{-i\omega\tau}] = 2iW(\omega) \sin\left(\frac{\omega\tau}{2}\right) e^{-i\omega\tau/2}, \quad (2)$$

where $W(\omega)$ is the frequency spectrum of the source wavelet, $\omega = 2\pi f$ is the angular frequency, f is the frequency in Hz and i is the imaginary unit. The amplitude spectrum $|S(\omega)|$ is the modulus of the complex spectrum $S(\omega)$ and its maximum value is reached when $|\sin(\omega\tau/2)| = 1$, that is, when $\omega\tau/2 = (2k+1)\pi/2$, $k = 0, 1, 2, \dots$. Therefore, the first maximum is given by $2f\tau = 1$. Replacing τ by its physical meaning, that is, the two-way traveltime of a P -wave propagating with velocity v_p in the thin layer, the following relationship is obtained:

$$2f \left(\frac{2d}{v_p} \right) = 1, \quad (3)$$

or equivalently,

$$d = \frac{\lambda_p}{4}, \quad (4)$$

since $\lambda_p = v_p/f$. This result has been highlighted by numerous authors, notably in the physical modelling study by R. Tatalovic *et al.* (1988) regarding P -wave tuning effects in thin layers.

For very thin beds, that is, when $d \rightarrow 0$, $\sin(\omega\tau/2)$ can be approximated by $\omega\tau/2$ and then, eq. (2) becomes:

$$S(\omega) = i\omega W(\omega)\tau e^{-i\omega\tau/2} = 2i\omega W(\omega) \frac{d}{v_p} e^{-i\omega d/v_p}, \quad (5)$$

where we recognize, in $i\omega W(\omega)$, the frequency spectrum of the first derivative of $w(t)$, in agreement with the initial observation of M.B. Widess (1973). The other factors in eq. (5) are an amplitude scaling factor of the thin layer response, which is proportional to the layer thickness d , and a slight phase shift, also proportional to the layer thickness, both of which tend towards 0 as $d \rightarrow 0$.

Eqs (1) and (2), written for seismic waves, may also be applied to the EM interface responses generated on both sides of a thin slab. However, these signals obey different physical principles, and the layer boundaries play a different role in electrokinetics: rather than merely reflecting the incoming (seismic) waves, they must be considered as effective sources radiating EM energy (see S. Garambois & M. Dietrich 2002). Consequently, for EM waves, the time delay separating the two arrivals generated by the slab is no longer the two-way traveltime in the thin layer but rather the one-way traveltime, at P -wave speed, of the disturbance that will create two successive interface responses (EM traveltimes are negligible and therefore neglected). Eq. (3) then becomes:

$$2f \left(\frac{d}{v_p} \right) = 1, \quad (6)$$

which can be rewritten as,

$$d = \frac{\lambda_p}{2}, \quad (7)$$

thus highlighting a fundamental difference between the seismic and EM composite responses of thin layers.

2.2 Electrokinetic transfer function

The formulation proposed in S. Pride (1994) and S.R. Pride & M.W. Haartsen (1996) regarding seismoelectromagnetic effects taking place in porous media will be adopted by this study. It accounts for porous media whose pores are filled with an electrically conducting solution, which results in the electrokinetic coupling between seismic and electromagnetic fields.

For the seismoelectric case, and considering plane waves, the amount of mechanical energy being converted into electromagnetic fields is given by the dynamic (frequency-dependent) transfer functions (TF) (S. Garambois & M. Dietrich 2001; C. Bordes *et al.* 2015):

$$\begin{aligned} \mathbf{E}(\omega) &= \frac{i\omega\tilde{\rho}(\omega)L(\omega)\beta_L(\omega)}{\tilde{\epsilon}(\omega)}\mathbf{u}(\omega) = \frac{\tilde{\rho}(\omega)L(\omega)\beta_L(\omega)}{\tilde{\epsilon}(\omega)}\dot{\mathbf{u}}(\omega) \\ &= \frac{\tilde{\rho}(\omega)L(\omega)\beta_L(\omega)}{i\omega\tilde{\epsilon}(\omega)}\ddot{\mathbf{u}}(\omega), \end{aligned} \quad (8)$$

where $\omega = 2\pi f$ is the angular frequency, and where $\mathbf{E}$, $\mathbf{u}$, $\dot{\mathbf{u}}$ and $\ddot{\mathbf{u}}$ are, respectively, the coseismic electric field, the (average) solid particle displacement, velocity and acceleration. $\tilde{\rho}(\omega)$ is the effective density of the pore fluid defined by eq. (A1) in Appendix A. $L(\omega)$, as detailed in eq. (A7), is a first-order coefficient controlling all couplings between mechanical and electromagnetic fields (S. Pride 1994). $\beta_L(\omega)$ is the ratio between the relative fluid-to-solid displacement vector $\mathbf{w}$ and the solid frame displacement $\mathbf{u}_{solid}$, defined by $\mathbf{w} \equiv \beta_L(\omega)\mathbf{u} \equiv \phi(\mathbf{u}_{fluid} - \mathbf{u}_{solid})$, where $\mathbf{u}_{fluid}$ is the fluid displacement and ϕ the porosity of the medium. $\tilde{\epsilon}(\omega)$ is the effective electric permittivity detailed in eq. (A4).

All physical parameters involved in eq. (8) are discussed and detailed in Appendix A.

3 EXPERIMENTAL AND NUMERICAL METHODOLOGY

Details about the experiments and the numerical simulations can be found in V. Martins-Gomes (2022) and in V. Martins-Gomes *et al.* (2023). Our experimental setup is depicted in Fig. 1. For the numerical simulations, we follow our previous procedure and that presented in N. Bernardo *et al.* (2024) to adapt the numerical code of S. Garambois & M. Dietrich (2002) and ensure that our simulations correctly represent the experiments (i.e. by accounting, as much as possible, for the large dimensions of the seismic source and the length of the array of electric receivers relative to the scale of the experiment). In what follows we detail the experimental and numerical methodology corresponding to the results discussed in Sections 4 and 5.

As depicted in Fig. 1, the whole experiment consists of a $24.7 \times 11.5 \times 11$ cm plastic container filled with water-saturated homogeneous sand. A water-saturated sandstone layer is inserted near the middle of the sandbox to study the EM interface response created at the sand–sandstone (interface I) and at the sandstone–sand (interface II) boundaries. The sand and the sandstone materials used in the experiments are the same as those employed in the works of V. Martins-Gomes *et al.* (2023) and N. Bernardo *et al.* (2024). Before and after the experiments, the sandstone layers were kept inside a container filled with distilled water that was periodically controlled to ensure that the layers are fully saturated and that the fluid conductivity remains in the range $[1.2\text{--}1.8]$ mS m^{−1}. For the experiments detailed in Section 5, the pore fluid conductivity of the sand was

adjusted using a NaCl solution; during measurements, the sandstone layer was sealed within a very thin ($d \ll 1$ mm) impermeable plastic bag to prevent osmotic ionic migration from the sand to the sandstone. We conducted a series of benchmark measurements to confirm that the plastic bag did not substantially alter our results [see V. Martins-Gomes (2022), chapter 4].

A piezoelectric transducer with a 250 kHz dominant frequency was used as a seismic source, placed on the outside of one of the sides of the container, and excited by a half-sine burst. Electric potentials V_{exp} were measured using a comb of electrodes, introduced in the sand along the y-axis, and a reference electrode placed in a second plastic container. Both containers are connected by a wire equipped with a 10 MΩ resistance. V_{exp} is the difference between the electric potential of an electrode in the comb and the reference. Because the second container is isolated (i.e. there are no SE signals, only EM noise), V_{exp} corresponds approximately to the electric potential measured by the electrodes in the comb. The electric resistance of the wire connecting both containers was empirically chosen and as long as it is significantly smaller than the 100 MΩ input impedance of the preamplifier used, it will not impact the measurements [see V. Martins-Gomes (2022), chapter 4] for a detailed analysis of the electronic circuit representing the SE measurements. Note that a single-point grounding configuration was used, with the oscilloscope serving as the earth reference. The seismic displacement perpendicular to the sand surface u_y was measured using a laser vibrometer (M.S. Devi *et al.* 2018; C. Shen *et al.* 2022) and a small 4 mm² retro-reflective paper placed on top of the sand to increase the signal-to-noise ratio. Both electric and seismic measurements were made along a z-oriented line, centred around the x-axis ($x \simeq 0$), and covering the ranges $z = [1\text{--}10.75]$ cm and $z = [13\text{--}22.75]$, adapted depending on the thickness of the sandstone layer used. The spacing between measuring points for the electric and seismic acquisitions are, respectively, 0.25 and 0.125 cm.

Regarding the numerical modelling in Section 4 and Section 5, we have used an explosive source and a three layer configuration to represent the experiment: the first and third layers correspond to the sand and the second to the sandstone. The physical properties considered here are the same as those already used by V. Martins-Gomes *et al.* (2023), unless otherwise specified. They are presented in Appendix B.

In the experiments discussed in Section 4, for each layer thickness a different sandstone sample, extracted from the same rock formation, was used and their physical properties are shown in Tables B1 and 1. The fluid properties of sand and sandstone was the same for the experiments presented in that section. Concerning Section 5, the experiments used the same 0.74 cm thick sandstone layer from Section 4, with pore fluid properties unchanged. Only the σ_f of the sand varied in the experiments discussed in Section 5 and Table 2 shows the experimentally measured values.

Sandstone layers were carefully cut and chosen to ensure their similarity. Nevertheless, differences in mechanical properties exist, leading to slightly distinct seismic velocities. For this reason, in order to compare experimental results and numerical simulations in Section 4, we have followed the methodology explained in Appendix C to determine the frame moduli (K_{fr} and G_{fr}) in our simulations that fit the *P*-wave velocities obtained from the experimental data. Table 1 displays the values of frame moduli and fluid conductivity used in the numerical simulations presented in Section 4.

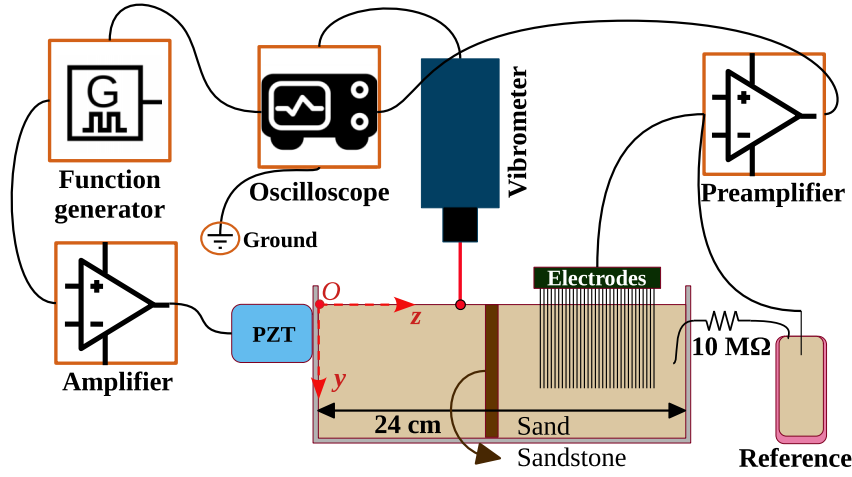


Figure 1. Sketch highlighting the main components of the experimental set-up as viewed from the lateral of the sandbox. The Cartesian coordinate system (x, y, z) has its origin O at the top of the sand, centred with respect to the x -axis (oriented perpendicular to the page) and corresponding in space to the topmost point of the PZT seismic source. The laser beam and the electrodes are oriented along the y -axis.

Table 1. Main physical parameters of the three-layer model used in the numerical simulations of Section 4. **I1**, **I2** and **I3**, refer, respectively, to the 1st, 2nd and 3rd layers.

Sandstone thickness (cm)		0.2	0.5	0.74	0.9	1.63	1.91	2.5	5
Sandstone z -limits (cm)		[11.98, 12.18]	[11.65, 12.15]	[11.88, 12.62]	[11.61, 12.51]	[11.6, 13.23]	[11.96, 13.87]	[11.8, 14.3]	[11.75, 16.75]
Bulk (frame) modulus (K_f)(Pa)	I1	2.38×10^7	2.89×10^8	2.38×10^7	2.38×10^7	1.06×10^8	2.44×10^8	2.38×10^7	2.38×10^7
	I2	3.41×10^8	3.41×10^8	3.41×10^8	3.41×10^8	3.41×10^8	2.1×10^9	3.41×10^8	1.83×10^9
	I3	2.38×10^7	2.89×10^8	2.38×10^7	2.38×10^7	1.06×10^8	1.47×10^8	2.38×10^7	2.38×10^7
Shear (frame) modulus (G_f)(Pa)	I1	1.43×10^7	1.74×10^8	1.43×10^7	1.43×10^7	6.33×10^7	1.47×10^8	1.43×10^7	1.43×10^7
	I2	2.1×10^8	2.1×10^8	2.1×10^8	2.1×10^8	2.1×10^8	1.33×10^9	2.1×10^8	1.16×10^9
	I3	1.43×10^7	1.74×10^8	1.43×10^7	1.43×10^7	6.33×10^7	8.81×10^7	1.43×10^7	1.43×10^7
P -wave velocity (v_p) (m s^{-1})	I1	1780	1820	1780	1780	1790	1810	1780	1780
	I2	2200	2200	2200	2200	2200	2500	2200	2460
	I3	1780	1820	1780	1780	1790	1800	1780	1780
Fluid conductivity (σ_f)(mS m^{-1})	I1	1.8	1.8	1.8	1.31	1.8	1.31	1.8	1.8
	I2	1.31	1.31	1.31	1.31	1.31	1.8	1.8	1.8
	I3	1.8	1.8	1.8	1.31	1.8	1.8	1.8	1.8

Table 2. Fluid conductivity and zeta potential of the sand in the experiments where the conductivity of the fluid, filling the pores of the sand, vary. The fluid conductivity was measured. For completeness, we also show the position along the z -axis of the sandstone layer used in Section 5.

Fluid conductivity of sand (mS m^{-1})	1.31	2	3.65	7.6	15.3
ζ (mV)	11.51	1	1.15	1	1.76
z -limits (cm)	[11.88, 12.62]	[11.7, 12.44]	[11.6, 12.34]	[11.62, 12.36]	[11.82, 12.56]

For the numerical simulations shown in Section 5, we have carried out a numerical sensitivity study by only varying the zeta potential of the sand layers in order to invert for ζ -values respecting the measured ratios between the IR and the coseismic amplitudes for each fluid conductivity. This strategy was adopted because neither ζ nor the fluid composition were measured. Additionally, a thorough study of the zeta potential is beyond the scope of this work, given that the dependence of ζ on the fluid conductivity in a porous medium is still a subject of active research (T. Ishido & H. Mizutani 1981; S.R. Pride & F.D. Morgan 1991; A. Revil *et al.* 1999; B. Lorne *et al.* 1999b; X. Guichet *et al.* 2003; A. Dukhin *et al.* 2005; X. Guichet *et al.* 2006; J. Vinogradov *et al.* 2010; E. Walker & P. Glover 2018; J. Vinogradov *et al.* 2018; S. Li *et al.* 2018). Note that for $\sigma_f = 1.31 \text{ mS m}^{-1}$, we used the same value of zeta potential as in V. Martins-Gomes *et al.* (2023), since for this σ_f -value our experiment is largely similar to our previous study. The values of the electric conductivity of the fluid in the sand, measured in the experiment, and the zeta

potential values used in the numerical simulations are summarized in Table 2.

Fig. 2 shows an example of the experimental and numerically simulated data for a model with a 2.5 cm thick sandstone layer inside the sand, and with the same fluid conductivity in both sand and sandstone materials (see Table 1 for details about this model). In the seismic displacement u_y records, we observe both the direct P -wave highlighted by the dashed magenta line and the P -waves reflected on the two sand-sandstone interfaces, indicated by the dashed blue lines. Their accompanying coseismic fields, highlighted in the same manner, can be observed in the electric potential V_{exp} records. The synchronous EM interface response (IR) created at both interfaces can also be observed in the electric data (dashed orange lines in Figs 2a and b). These IRs at z -offsets smaller than the position of the first interface are less pronounced in the numerical records than in the experimental data, possibly due to 3-D wave phenomena that cannot be accounted for in the simulations, as extensively discussed in V.

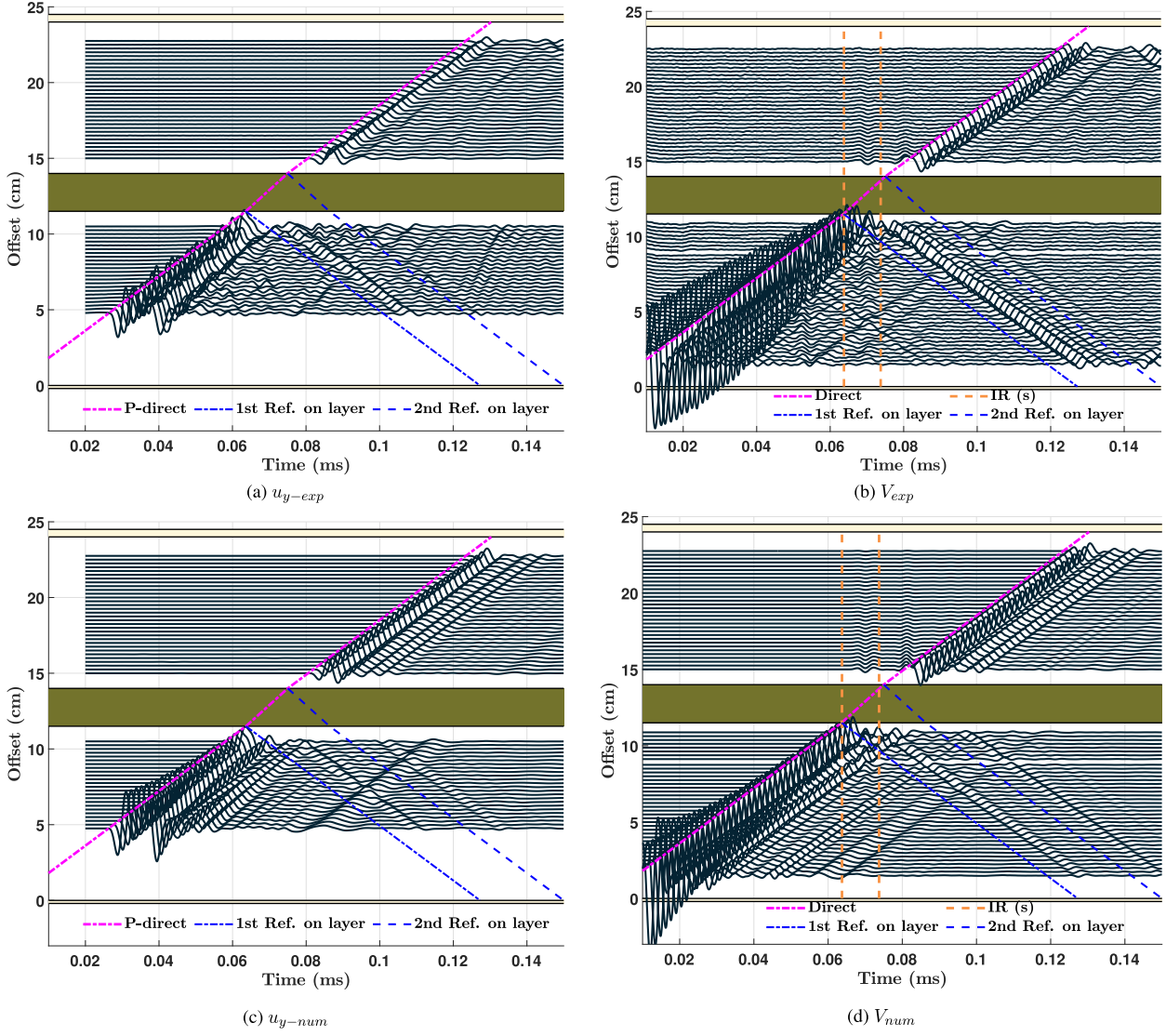


Figure 2. Experimental and simulated seismoelectric responses of a model containing a 2.5 cm thick sandstone layer in the middle of the sandbox, as a function of time and source-receiver offset. Refer to Table 1 for the physical parameters. (a) and (c) show the particle displacement: $u_{y\text{-exp}}$ and $u_{y\text{-num}}$. (b) and (d) show the electric potential: V_{exp} and V_{num} . The front and back walls of the sandbox, and the sandstone layer are shown for convenience. The direct wave emitted by the seismic source and its coseismic signature are indicated by the magenta dashed lines. The P -wave (and coseismic) reflections on the sand–sandstone interfaces are highlighted by the dashed blue lines. The two orange dashed lines in (b) and (d) indicate the EM interface responses (IR) created at the sand–sandstone interfaces.

Martins-Gomes *et al.* (2023). Overall, there is a good agreement between the numerical simulations and the experimental data.

4 EFFECT OF VARYING THE SANDSTONE THICKNESS

To illustrate how thickness d modifies the IR and the reflected P waves, we have chosen to focus on the measurements at offset $z = 18.5$ cm for the IR, and at $z = 8.25$ cm for the reflected P wave. In Fig. 3 we show, for each thickness d used in the experiments, the band-passed (150 kHz–500 kHz) experimental records after detrending and their corresponding numerical simulations, obtained with the parameters given in Table 1 and the methodology explained in Section 3. The vertical axis corresponds to d/λ_P values, considering $\lambda_P = 1$ cm in the sandstone

layer (based on experiments). Note that the time axes were modified to start at the onset of the IR in Fig. 3(a) and at the onset of the reflected P -wave signals in Fig. 3(b). The onset times were obtained from the layer position and the experimental P -wave velocity. Because the effect of d on reflected P waves is already well understood (E. Carabelli & R. Folicaldi 1957; M.B. Widess 1973; N. de Voogd & H. den Rooijen 1983; R. Tatalovic *et al.* 1988; Y. Liu & D.R. Schmitt 2003; G. Papageorgiou & M. Chapman 2021), in what follows we give special attention to how the IR evolves with layer thickening.

The V_{exp} data in Fig. 3(a) summarizes the dependence of the IR signal on d . When the thickness decreases, the IR created at II (sandstone–sand interface) approaches the IR signal created at I (sand–sandstone interface). For d/λ_P equal to 2.5 and 5, the two IR signals are easily distinguished, with the amplitudes of the second IR being smaller than that of the first one as a result

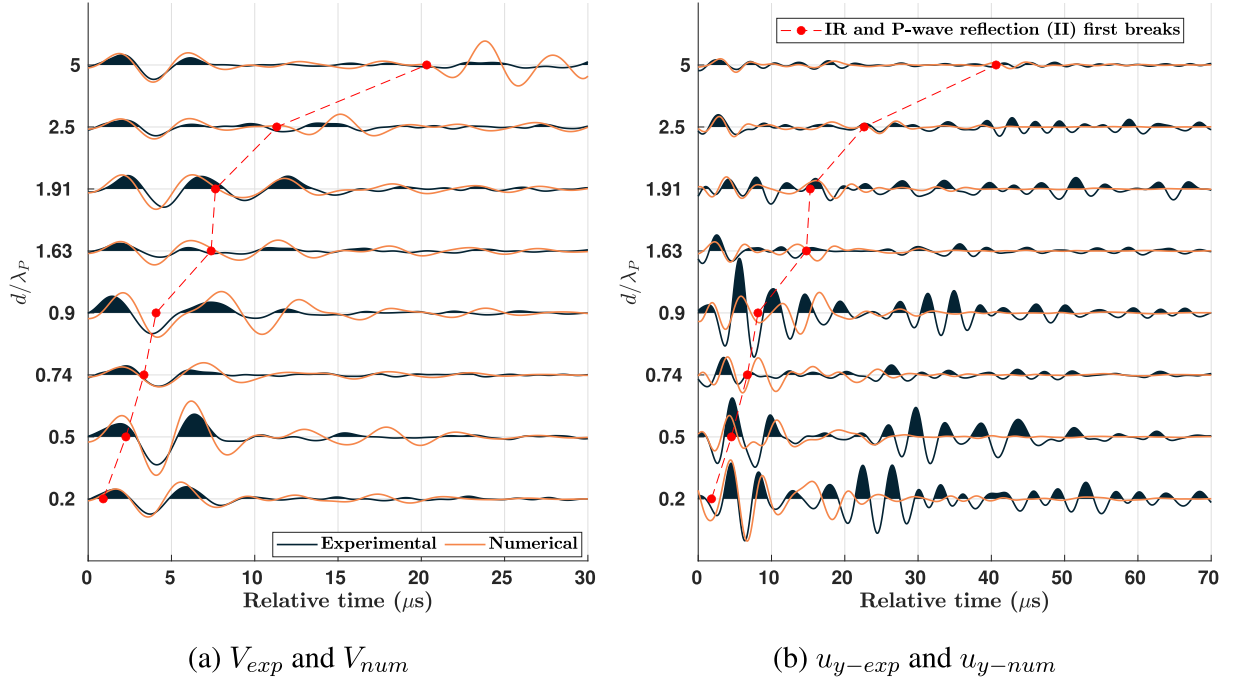


Figure 3. Comparison between experimental (dark) and numerical (orange) data for different thicknesses d . Note the difference in timescales. (a) IR signals at $z = 18.5$ cm. Considering the P -wave velocities in Table 1, (a) does not contain coseismic signals. (b) u_{y-exp} and u_{y-num} at $z = 8.25$ cm. The vertical axes indicate the value d/λ_P , with $\lambda_P = 1$ cm. The horizontal axes show the relative times based on the onset of the IR (in a), and of the reflected P wave (in b), created at interface I. The dot-dashed red lines indicate the first break, based on the experimental P -wave velocities, of the IR (in a) and of the reflected P wave (in b) created at interface II.

of P -wave energy partitioning at interface I and its attenuation inside the sandstone slab, and due to the IR attenuation from interface II to the electrode. The intrinsic attenuation of the second IR may also be due to rock composition heterogeneities and/or partial saturation inside the sandstone layer.

In the u_{y-exp} records shown in Fig. 3(b), layer thinning promotes interference between reflections from interfaces I and II, causing the reflected P wave and its amplitude to vary considerably with d . We also observe that secondary reflections, due to waves that first hit the plastic box and then the sandstone, have more energy in the experimental measurements obtained using thinner layers (events after 10 μs in Fig. 3b). Thickening of the sandstones slabs alters the wave paths in the experiments, and impacts these secondary waves in the sense that they will spend more time and further attenuate inside thicker layers.

A comparison of the numerical and experimental data in Fig. 3 shows that the simulations accurately reproduce both the waveforms and the arrival times of the signals measured in the sand-box, for all layer thicknesses. Notably, for the 5 cm thick layer the comparison in Fig. 3(a) shows an amplitude mismatch for the IR created at interface II, likely due to heterogeneities, not considered in the modelling, inside the layers. Regarding the u_{y-num} results in Fig. 3(b), we note an amplitude enhancement at $d/\lambda_P = 0.25$, in agreement with the findings of R. Tatalovic *et al.* (1988) and with eq. (4).

M.B. Widess (1973) depicts his results regarding the effect of thickness d on reflected seismic waveforms in his Fig. 2, which are essentially equivalent to our results in Fig. 3. The resemblance between the IR results and those obtained for reflected P waves comes from the fact that the waveforms of the IR are controlled by the time-dependent electric charge imbalance across

an interface, which is itself a function of the seismic energy passing through it (M.W. Haartsen & S.R. Pride 1997).

Fig. 4 shows the normalized magnitude spectra of experimental and numerical data for the time windows shown in Fig. 3. In the electric potential spectra of Fig. 4(a) we observe a fairly high degree of similarity between numerical and experimental spectra for all thicknesses. On the contrary, in Fig. 4(b) the experimental and numerical seismic displacement u_y spectra differ due to frequency components associated with complex 3-D wave propagation phenomena in the experiments that are not accounted for in the simulations, notably the P -wave reflections in the plastic box walls, as already observed in Fig. 3(b).

To further analyse our results, we conducted full-waveform and simplified numerical simulations. The simplified modelling consists in combining the wavelet associated with the first IR created at interface I with a time-delayed and amplitude-reversed copy of that wavelet, representing the IR generated at interface II, to predict the global IR response of an ideal non-attenuating medium at normal incidence. The reference wavelet chosen for these operations is taken from the IR generated at interface I for the 5 cm thick layer in the numerical simulation (i.e. the event taking place between 0 and 10 μs in Fig. 3), since for this thickness the IRs from both interfaces do not interfere. We assume that the P -wave velocity in the sandstone layer is 2200 m s⁻¹ and consider d -values ranging between 0.014 and 5 cm (with a 0.014 cm increment). The full-waveform numerical modelling calculates the IR signals detected by an electrode positioned at $z = 18.5$ cm, with the methodology described in V. Martins-Gomes *et al.* (2023), and using the parameters of the 2.5 cm thick layer in Table 1 to simulate a layer starting at $z = 11.75$ cm and with thicknesses varying in the same

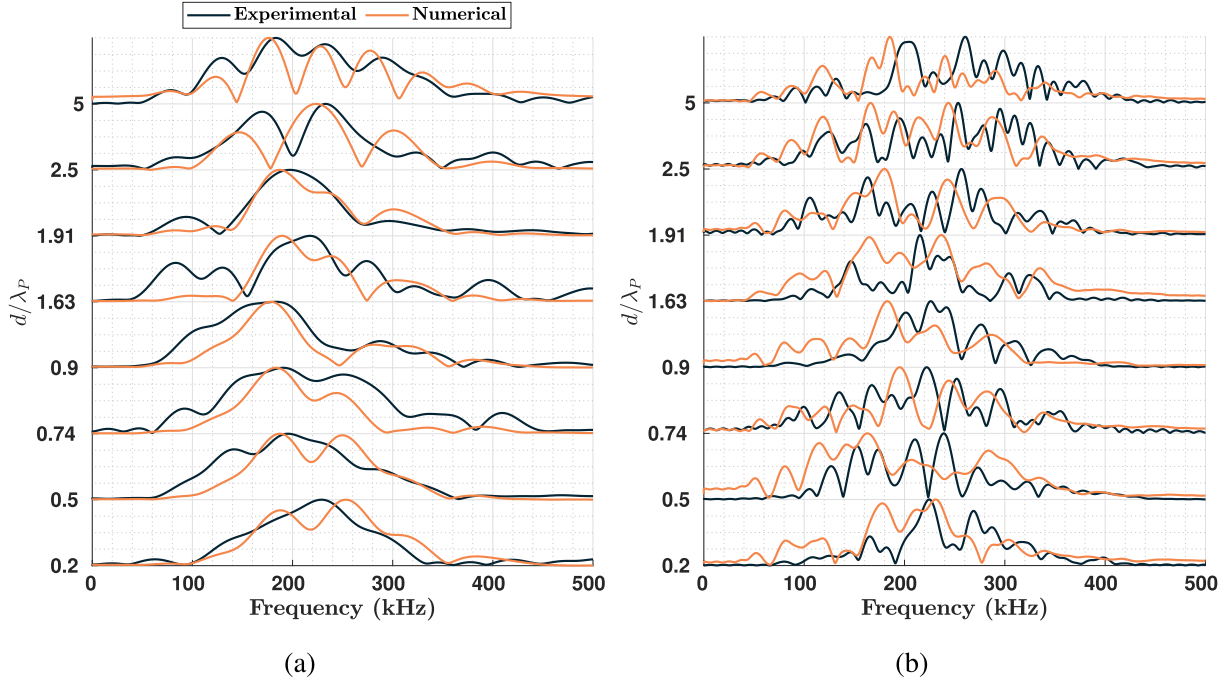


Figure 4. Normalized magnitude spectra of traces in Fig. 3. (a) corresponds to the spectra of the electric potential V data, and (b) corresponds to the spectra of the seismic displacement u_y data.

range as those taken into account in the simplified modelling. Simplified and full numerical traces, as well as their magnitude spectra, are depicted in Fig. 5, where we observe a striking similarity. This similarity tends to show that internal multiples within the sandstone layer are almost negligible. Additionally, we remark that the differences between the results of both modelling strategies increase with d because attenuation effects have been neglected in the simplified case. For the sake of completeness, the wavelet used in the simplified computations as well as its derivative (properly amplitude normalized) are also shown.

The full numerical and simplified modelling traces in Fig. 5(a) highlight how the IRs created at both interfaces will interfere with layer thinning, evolving towards a signal whose waveform is the derivative of the IR wavelet created at interface I, as observed by M.B. Widess (1973) for reflected P waves and shown by eq. (5). Moreover, the behaviour shown by the IR waveforms as a function of d/λ_P closely reflects the results of Fig. 3(a).

Fig. 5(b) shows the normalized magnitude spectra of the traces in Fig. 5(a). The computed spectra vary considerably with layer thickening, evolving from a band-pass filter when $d < \lambda_P$ to a multifrequency notch filter when $d > \lambda_P$ (assuming $v_P = 2200 \text{ m s}^{-1}$ and a dominant frequency of 210 kHz). The spectrum of the resulting IR is consistent with our observations made from the traces in Fig. 5(a): as $d \rightarrow 0$ it converges towards the spectrum of the derivative of the IR created at interface I (dashed red line in Fig. 5b). Note that the variation of the spectra in Fig. 5(b) with d is similar to that shown in Fig. 4(a). J. Deparis & S. Garambois (2009) have also observed the same magnitude spectra variation when studying the effect of thin-beds on the EM reflection coefficients in the context of ground-penetrating radar exploration.

In order to complement the previous discussion, the first maxima and minima of the experimental, full numerical and

simplified IR waveforms (including contributions from both interfaces of the layer) are shown in Fig. 6 as a function of d/λ_P and d . In this figure, three d/λ_P values have been highlighted based on the curves obtained from the simplified modelling. Considering $\lambda_P \simeq 1 \text{ cm}$ in the sandstone layer (based on the experiments), the maximum reaches a constant when $d/\lambda_P > 0.3$ (0.3 cm), indicating the d -values for which the second IR no longer affects the first maximum. Regarding the minima, the same behaviour exists for $d/\lambda_P > 0.78$ (0.78 cm), approximately, where the two IRs stop interfering. Around $d/\lambda_P = 0.5$ ($\simeq 0.45 \text{ cm}$), the minima display rapid amplitude changes, with a peak absolute value that is nearly 1.7 times larger than the amplitude associated with a single interface. Note that these values strongly depend on the characteristics of the wavelet used to calculate the simplified response, namely the dominant frequency, bandwidth and shape of the wavelet. At last, since curves associated with the simplified modelling and full-waveform modelling are very similar in Fig. 6, by extension, the discussion above is also valid for the full-waveform approach.

By comparing the experimental data and those obtained from the simplified modelling in Fig. 6, one can also note a good agreement with the first maxima and minima of the experimental data, following the trend (blue dots in Fig. 6) imposed by the simplified modelling values. The results in Fig. 6 can be readily compared with those of R. Tatalovic *et al.* (1988). The simplified modelling trends agree almost perfectly with those in fig. 9 of R. Tatalovic *et al.* (1988), and the behaviour of our experimental minima is arguably similar as well, even though the lack of data from more thickness values limits the comparison.

The results presented above confirm that the relationship between the IR amplitude and thickness d strictly follows what has already been observed for reflected seismic waves and reported in the literature. Moreover we have found that, as long as only d

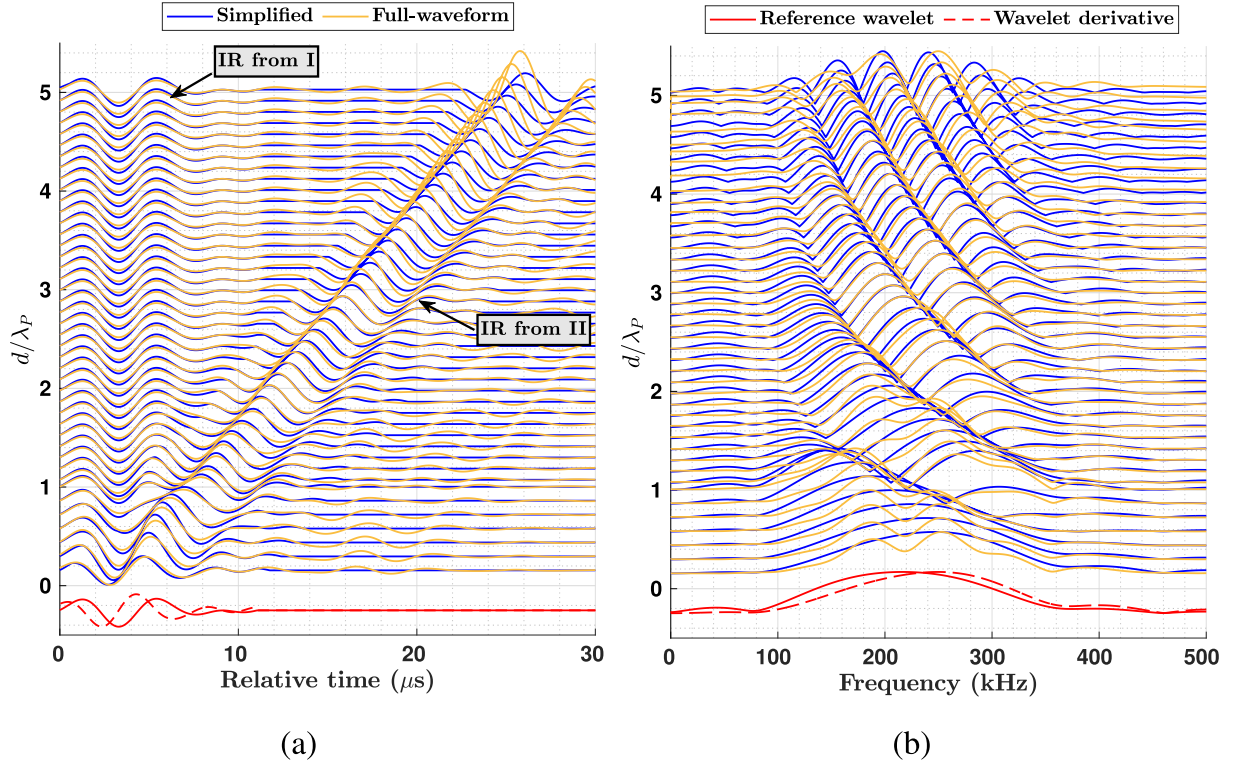


Figure 5. (a) IRs obtained from simplified (blue) and full numerical (orange) simulations, considering $z = 18.5$ cm. The simplified modelling uses the wavelet (shown as the red solid line) extracted from the numerical simulation made for a 5 cm thick layer in Fig. 3(a). The dashed red signal is the derivative of that wavelet, normalized such that it has the same energy as the original wavelet. The origin of the time axis corresponds to the arrival of IR signals from interface I. The vertical axis uses $\lambda_P = 1$ cm. (b) Normalized magnitude spectra of the traces shown in (a).

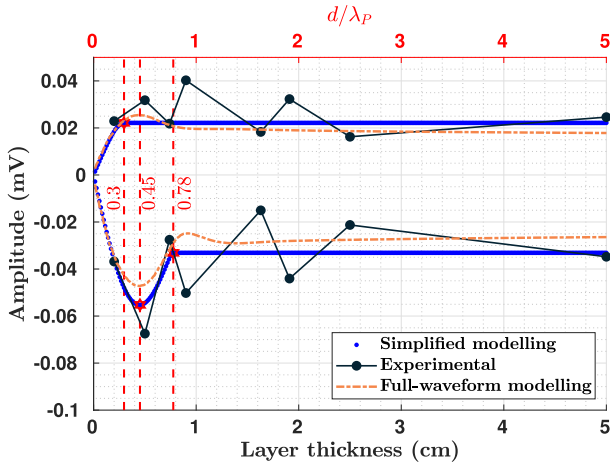


Figure 6. First IR maxima and minima derived from the experimental, full numerical and simplified modelling approaches as a function of thickness d at the z -offset of 18.5 cm. The upper horizontal axis uses $\lambda_P = 1$ cm, and the vertical red lines highlight d/λ_P -values where the amplitude curves show significant changes.

matters, an amplification of IR signals occurs when $d/\lambda_P \simeq 0.5$, in agreement with eq. (7). Despite that, the total amplitude increase is not so significant since the enhanced amplitudes have the same order of magnitude as the IR created by a single layer. These results are in agreement with the numerical observations of S.S. Haines & S.R. Pride (2006), but disagree with the conclusions of N. Grobbe & E.C. Slob (2016), also based on numerical

simulations, that reported a maximum amplification for thicknesses close to the dominant wavelength λ_P . However, it should be recalled that they focused their study on the effect of different stacks of thin layers. There is also a disagreement with the experimental results of R. Peng *et al.* (2017), that found, considering $\lambda_P = 1$ cm, that amplitudes always increase with layer thinning. Finally, the results of Y. Kang *et al.* (2024) point to a decrease of IR amplitudes with layer thinning when $d < 0.1\lambda_P$, in agreement with what we have observed here and with the predictions of eq. (5).

5 EFFECT OF VARYING THE FLUID CONDUCTIVITY

In order to further explore the effect of a thin layer on the IR amplitude and waveform, we focus here on how a contrast of fluid conductivity between sand and sandstone media impacts this signal. Using the methodology outlined in Section 3 and considering the same generic three-layer model (sand-sandstone-sand) as before, we provide new experimental data and compare it with numerical simulations using Pride's theory. Fig. 7 shows the electric potential measurements obtained with a 0.74 cm thick sandstone layer and five different values of fluid conductivities in the sand, ranging from 1.31 to 15.3 mS m⁻¹. The σ_f of the sandstone did not change throughout the experiments discussed in this section and its value, shown in Table 1, was 1.31 mS m⁻¹. Although we use the electric acquisition geometry described in Section 3, here we only analyse z -offsets ranging from

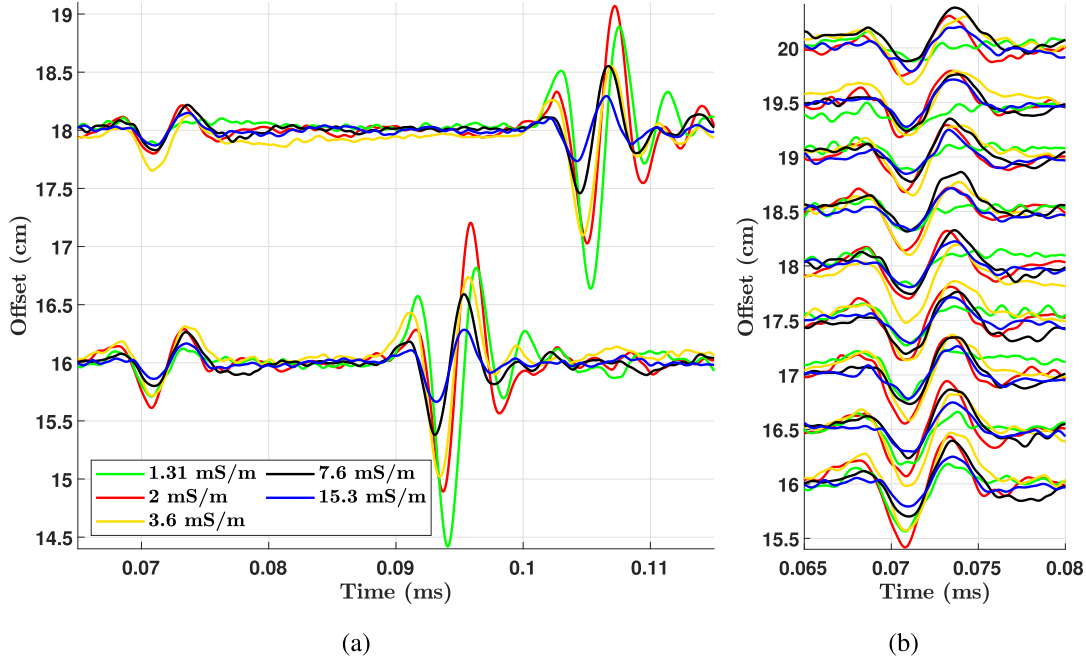


Figure 7. Electric potential traces V_{exp} obtained by varying the fluid conductivity in the sand when a 0.74 cm thick sandstone layer was used. (a) EM interface response (earlier event) and coseismic arrivals measured at two offsets along the acquisition line. (b) Interface response at 9 offsets, showing how its waveform and amplitude change in the experiment.

13 to 23 cm, where the IR and coseismic signals can be easily distinguished. A band-pass filter ([50 kHz–1 MHz]) was used to attenuate low and high-frequency noises, and a detrending procedure was applied to remove increasing/decreasing linear trends in the amplitudes of the measured signal.

In the two traces displayed in Fig. 7(a), the synchronous event before 0.08 ms is the total IR of the sandstone layer, and the arrivals after 0.09 ms correspond to the coseismic waves. From the coseismic signals it is evident that the amplitudes decrease when increasing the fluid conductivity in the sand. In Fig. 7(b), the IR measured at nine different offsets is shown in order to highlight how it decreases with increasing offset and how it evolves with changes in fluid conductivity. Unlike the coseismic signals, the amplitudes of the IR show a different response to an increase of σ_f in the sand. In most z -offsets, the highest IR amplitude occurs when the fluid conductivity is equal to 3.6 mS m^{-1} , implying that for lower σ_f values the amplitude of IR signals will increase as the sand’s fluid conductivity rises.

In order to further analyse our experimental data, we have compared them to numerical simulations, as detailed in Section 3. We have taken the physical parameters already used in V. Martins-Gomes *et al.* (2023), except for the ζ and σ_f values of the sand, which are shown in Table 2. The superimposition of the experimental electric potentials V_{exp} and the numerically calculated electric potentials V_{num} is shown in Fig. 8 for two offsets. Fig. 9 shows a comparison between both data sets in terms of the first minima and maxima of the coseismic signal, and in terms of the absolute maxima of the interface response. Table 3 displays the scaling factors by which the V_{num} signals have been multiplied to superimpose them with V_{exp} in Fig. 8. Note that the comparisons in Figs 8 and 9 are particularly dependent on the ζ value of the sand since we inverted for this parameter, as explained in Section 3.

Fig. 8 shows a good agreement between V_{num} and V_{exp} : both the coseismic signals (waveforms at times greater than 0.09 ms) and the EM interface response (around 0.07 ms) are satisfactorily simulated, since the predicted waveforms are in close agreement with those obtained from experiments (Fig. 7). In our comparison, the highest scaling factor (i.e. the furthest from 1) corresponds to the lowest σ_f value used (Table 3), in agreement with the observations made by G.I. Block & J.G. Harris (2006) and M.D. Schakel *et al.* (2011) that higher σ_f values favour the comparison between experimental data and numerical simulations based on Pride’s theory.

Fig. 9 shows the first maxima and minima of the displacement (u_y) measurements on top of the sand, of the coseismic measurements (V_{exp}), and of the absolute maxima of the interface response ($V_{\text{EM-exp}}$). The amplitudes obtained from numerical simulations are also displayed in Fig. 9 after applying the scaling factors given in Table 3, providing a global view of how the theoretically predicted amplitudes of the coseismic signals and of the IR compare with the amplitudes derived from the experimental measurements.

Concerning the amplitudes of u_y , one can note in Fig. 9 that instead of being the same for all experiments (since it should only weakly depend on σ_f), it varies due to unavoidable fluctuations in the experimental conditions. The coseismic amplitudes in Fig. 9(b) decrease with increasing σ_f , and we note that this amplitude trend is not linear, as anticipated by the fact that ζ also varies with σ_f in a nonlinear manner (see S.R. Pride & F.D. Morgan 1991), and also because both (σ_f and ζ) are first-order parameters in the SE transfer function (eq. 8). The maximum absolute amplitudes of the IR are shown in Fig. 9(c) only for z -offsets greater than or equal to 16 cm, since at offsets z -offsets < 16 cm the interference between coseismic and IR signals strongly controls the experimental amplitudes. At the z -offsets shown in Fig. 9(c), amplitudes decay with σ_f , however, when $\sigma_f = 1.31$

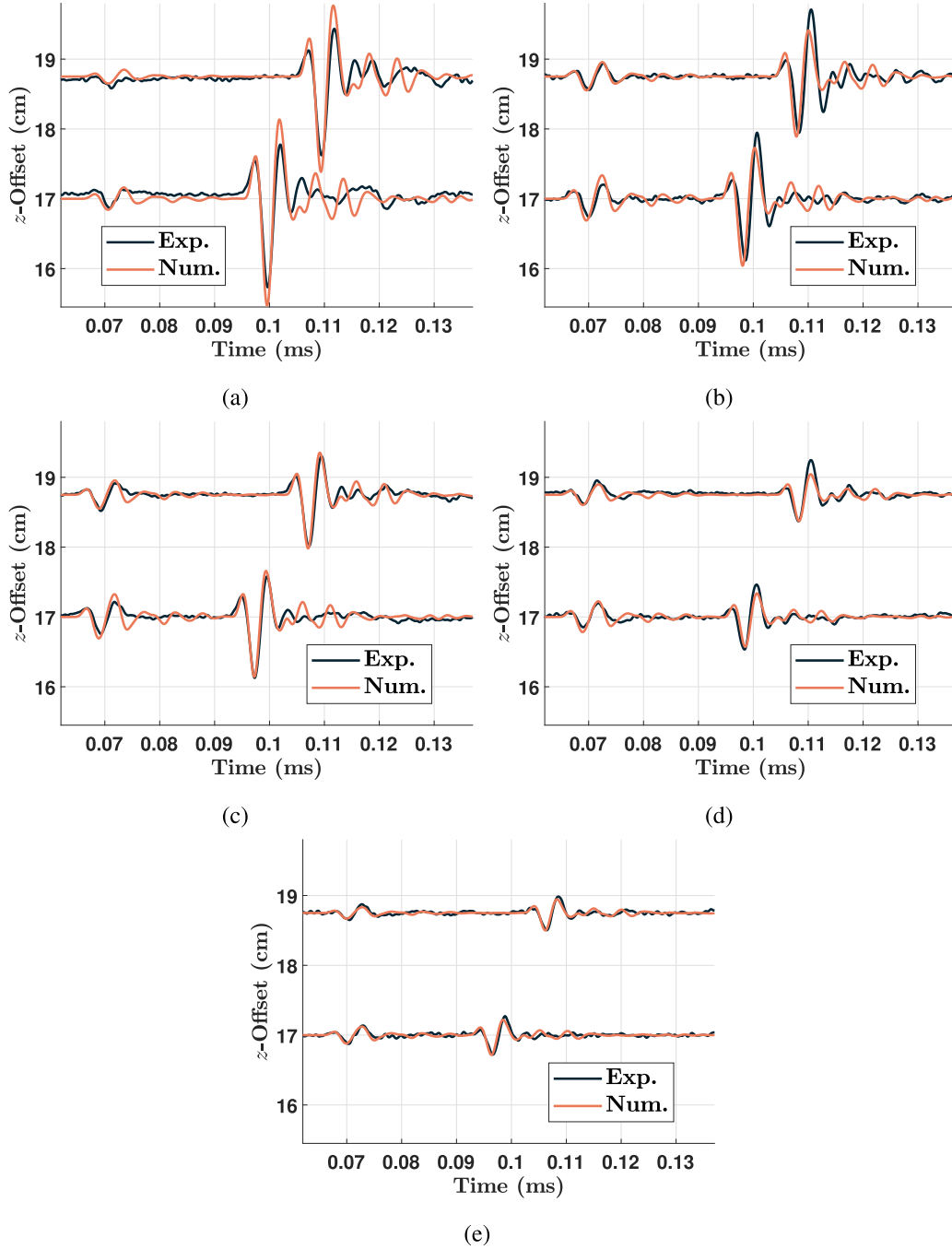


Figure 8. Superposition of V_{num} (orange) and V_{exp} (dark blue) signals at z -offsets 17 and 18.75 cm for various values of the fluid conductivity in the sand. (a) 1.31 mS m^{-1} , (b) 2 mS m^{-1} , (c) 3.6 mS m^{-1} , (d) 7.6 mS m^{-1} and (e) 15.3 mS m^{-1} . The scaling factors shown in Table 3 were used to fit V_{num} to V_{exp} .

mS m^{-1} (the lowest value used) the IR amplitudes are either lower or equal to the values obtained with other σ_f values, suggesting the existence of a complex σ_f -dependence, in contrast with what we observed for the coseismic amplitudes (Fig. 9b).

Regarding the comparison between numerical and experimental amplitudes in Fig. 9, the simulations provide a fair estimation of first maxima and minima, with more consistency at $z > 16 \text{ cm}$, where the coseismic and the IR do not interfere. Concerning the absolute maxima of the IR (Fig. 9c), the experimental noise level increases amplitude uncertainty, limiting the

interpretation of how changes in fluid conductivity affect this signal. Despite that, our numerical simulations were able to reproduce the overall σ_f -dependence of the experimental IR amplitudes shown in Fig. 9(c).

Regarding previous studies on this subject, most of the experimental work on seismoelectric conversions report a decay of coseismic amplitudes when increasing the fluid conductivity (e.g. B. Chen & Y. Mu 2005; J. Holzhauser *et al.* 2017). However, there is less consensus regarding the IR. In M.D. Schakel *et al.* (2011), numerical and experimental data have been used to explore how

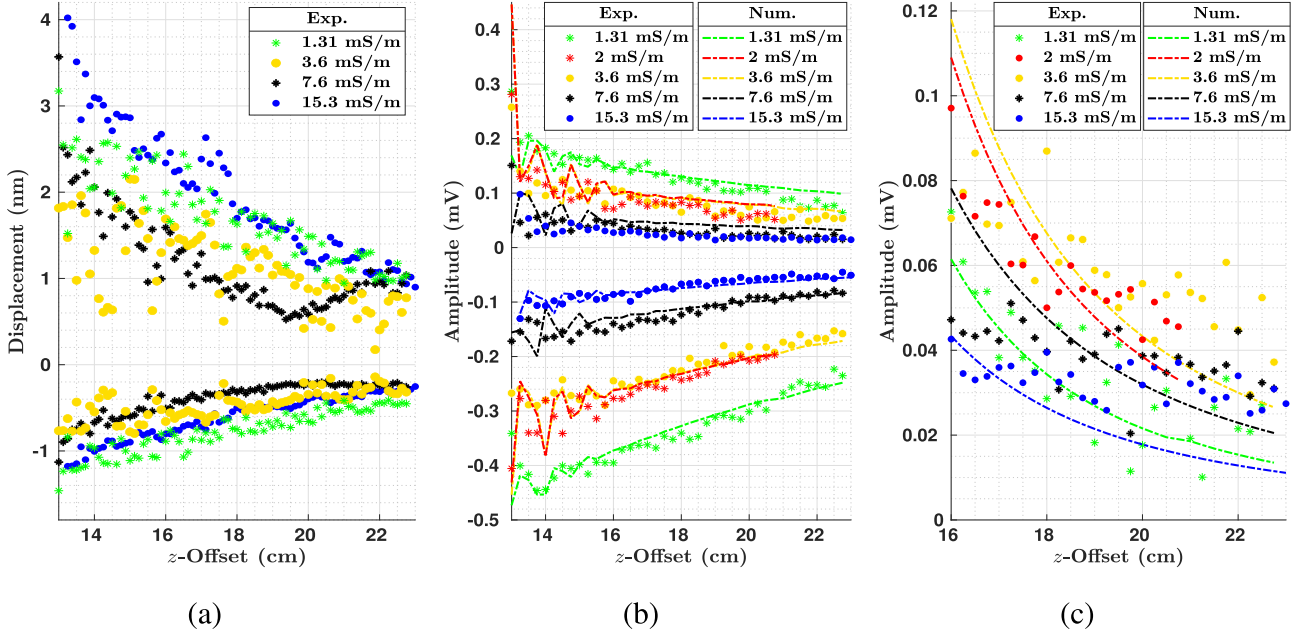


Figure 9. Experimental amplitudes acquired at various z -offsets and with different values of the fluid conductivity in the sand, using the 0.74 cm-thick sandstone layer. (a) first maxima and minima of u_y , (b) first maxima and minima of the coseismic response and (c) absolute maxima of the IR. In (b) and (c), the numerically calculated amplitudes are displayed as dashed lines by using the scaling factors given in Table 3.

Table 3. Values used to scale the numerical simulations for each conductivity tested in the sand.

Conductivity of sand (mS m^{-1})	1.31	2	3.5	7.6	15.3
Scaling factor	0.13	1.11	1.33	1.49	1.05

the amplitudes of the IR created at a fluid-porous media interface varies when σ_f in both media takes the following values: 1.27, 12 and 101 mS m^{-1} . Their figs 11, 12 and 13 show that IR amplitudes increase when σ_f goes from 1.27 to 12 mS m^{-1} but decrease with higher fluid conductivities until 101 mS m^{-1} , suggesting that amplitudes increase with σ_f until a certain value, after which they decay in proportion to the increase in σ_f . The same behaviour was reproduced numerically by G.I. Block & J.G. Harris (2006) and observed experimentally by Z. Zhu & M.N. Toksöz (2003). These observations are in agreement with the results we show in Fig. 9(c), although our study focus on σ_f contrasts at an interface between two porous media.

Fig. 10 shows how the ratio between the experimental amplitudes of the IR (Fig. 9c) and those of the first maxima and minima of the coseismic signal (Fig. 9b) vary with σ_f . We observe that the amplitude ratio is roughly proportional to σ_f . Because seismic-to-coseismic energy conversions are governed by eq. (8), there is an inverse proportionality between coseismic amplitudes and σ_f through $\sigma(\omega)$, as shown by eq. (A6). Conversely, any seismic-to-EM energy conversion creating IRs depends on the contrast of physical properties across the interface that it radiates from (see the sensitivity study in S. Garambois & M. Dietrich 2002), despite the fact that, in a porous medium, the attenuation of EM signals is proportional to σ_f . This essentially means that in the experiments presented here, increasing σ_f in the sand will further attenuate the IR signals measured, but will also lead to more P-to-EM wave conversions at the two interfaces. The competition between these two effects creates the IR

vs σ_f behaviour observed in Fig. 9(c) and Fig. 10. A similar observation was made by N. Bernardo *et al.* (2024) regarding the role of the dielectric permittivity on IR signals.

Finally, our experimental and numerical data lead to conclusions similar to those found in the literature. However, the fact that we treat the variation of coseismic and IR wavefields with σ_f simultaneously is a novelty that allowed us to successfully compare the predictions of Pride’s theory regarding these two SE fields in fluid-saturated layered porous media.

6 FIELD-SCALE NUMERICAL SIMULATIONS

In order to investigate how layer thickness d and σ_f contrasts may affect the IR in a field experiment, we sought to contextualize our previous results by performing numerical simulations in the appropriate frequency range. Fig. 11(a) shows the layout of the numerical simulations. We have considered an explosive point source, corresponding to a 6.6 kg dynamite explosion using the definition of M.W. Haartsen & S.R. Pride (1997), located at the origin of the xy -plane, at a depth of $z = 0.5$ m, and with a signature described by a 50 Hz peak-frequency Ricker wavelet. For seismic and electric acquisitions the receiver is located at offset $x = 50$ m, and buried at a depth of $z = 0.1$ m. The geological model consists of three layers: top and bottom layers are water-saturated sand with the same physical properties as the sand layers enclosing the 0.2 cm sandstone layer in Section 4 (i.e. layers 11 and 13 in Table 1). The middle layer is either the sandstone layer with properties of the 0.2 cm thick layer in Table 1 (layer 12) or a clay-like material whose physical properties are specified in Table 4. The thickness of the middle layer varies between 0.5 and 100 m, with a 1 m step. In the sandstone case, we have carried out simulations using five different $\sigma_{f\text{-sand}}$ values (fluid conductivity in top and bottom layers): 1.31, 3.6, 15.3, 110 and 1000 mS m^{-1} .

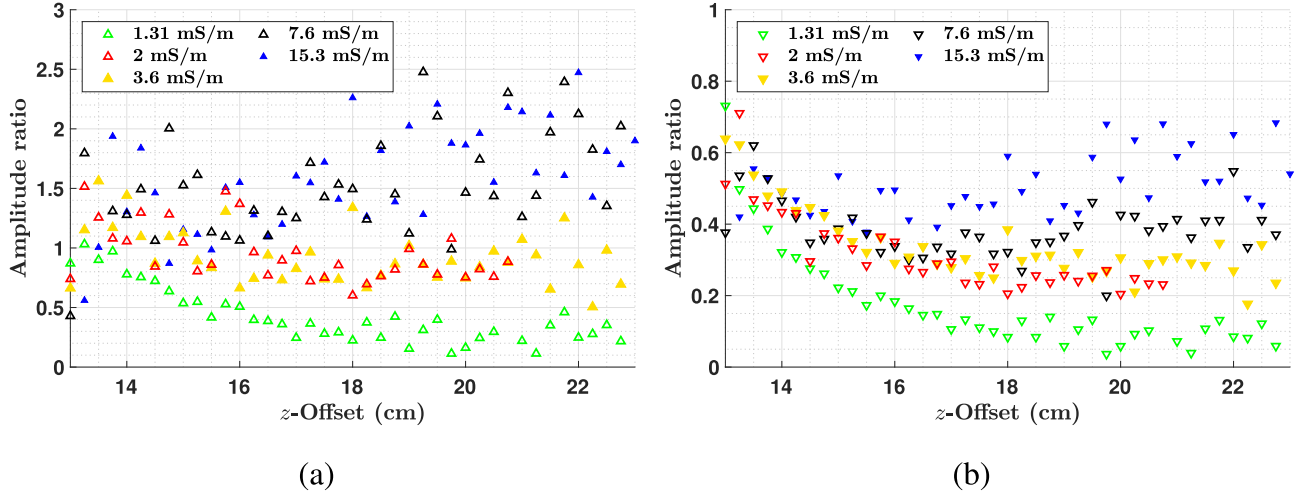


Figure 10. Experimental amplitude ratios between the absolute maxima of the EM interface response and the coseismic amplitudes due to P waves incident on a 0.74 cm thick sandstone layer, located inside a fully saturated homogeneous sand, for various values of the electrical conductivity of the saturating fluid. (a) with respect to the first maxima of the coseismic signals and (b) with respect to the first minima.

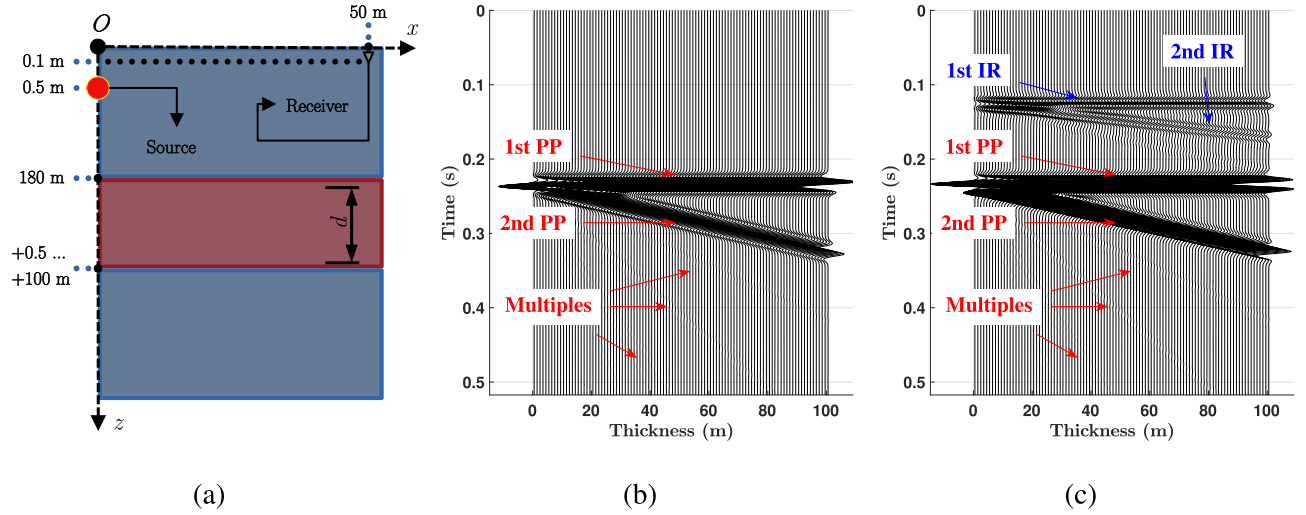


Figure 11. (a) Acquisition geometry used in the numerical simulations. Electric and seismic receivers are placed at $x = 50$ m, and at a depth $z = 0.1$ m. An explosive source is triggered at a depth of 0.5 m below the origin O . Top and bottom layers have the same properties. The middle layer starts at $z = 180$ m and its thickness d ranges from 0.5 to 100 m, with a 1 m step. (b) and (c) Seismic displacement u_z and electric potential V_{num} recorded when $\sigma_f = 1.31 \text{ mS m}^{-1}$ in all layers, with P-to-P (or coseismic) wave reflections and IRs indicated by arrows. In order to highlight the IRs, the direct waves have not been included in the simulations.

Table 4. Physical properties of the clay-like material used in the numerical simulations of Section 6. The other physical parameters are equal to those of the sandstone layer in Table B1.

ϕ	k_0 (m ²)	K_s (Pa)	K_{fr} (Pa)	G_{fr} (Pa)	ρ_s (kg m ⁻³)	C_0 (mmol l ⁻¹)	ζ (mV)
0.1	1×10^{-16}	19×10^9	6×10^9	7×10^9	2700	10	-44

In order to make our numerical simulations closer to field measurements, we have considered the electric potential V calculated using the 3-electrode approximation (M.S. Devi *et al.* 2018; M. Dietrich *et al.* 2018), with a 1 m spacing between electrodes, since it is more feasible in field surveys than the electrode configuration used in our laboratory experiments. Fig. 11 shows the z -oriented displacement (11b) and the approximate potential V (11c) recorded by the receiver (at $x = 50$ m) when all the layers of the model are characterized by $\sigma_f = 1.31 \text{ mS m}^{-1}$. Red

arrows in Figs 11(b) and (c) indicate P waves (and their corresponding coseismic arrivals) reflected at both interfaces, as well as 2nd-order events associated with multiple reflections inside the middle layer. The IRs created at both interfaces are indicated with blue arrows and, despite not being visible, 2nd-order IR signals created by P waves reflected back and forth inside the middle layer do exist.

Fig. 12 shows the first maxima and minima of the reflected P waves (Fig. 12a) and of the IRs (Fig. 12b) created using

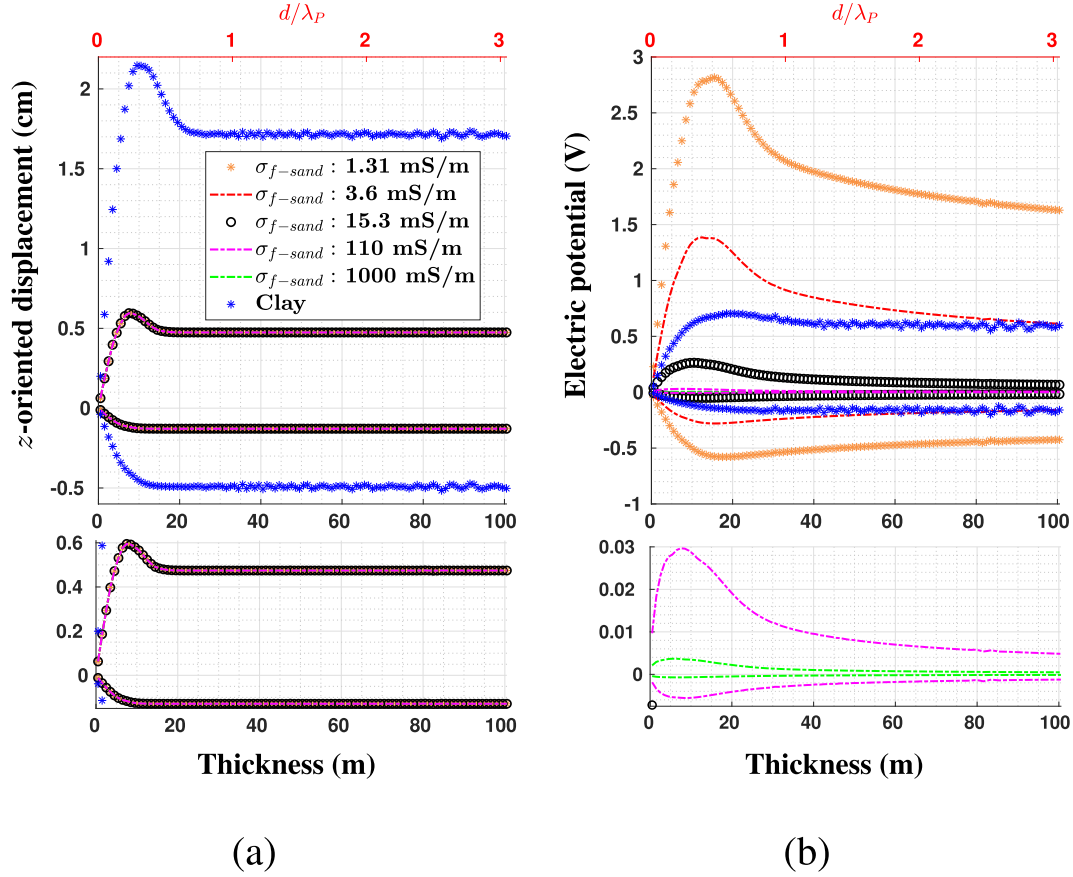


Figure 12. First maxima and minima of reflected P waves (a) and IRs (b) created by the three-layer model shown in Fig. 11(a). Except for the case where the clay-material of Table 4 is considered (blue asterisks), all other numerical simulations consider a sandstone material (properties of the 0.2 cm thick layer in Table 1) for the middle layer, differing only by the σ_f value used for the sand layers (top and bottom). When a clay layer is used, $\sigma_{f\text{-sand}} = 15.3 \text{ mS m}^{-1}$. The lowermost plot in (a) and (b) highlight lower values of u_z and V . The upper horizontal axis show d/λ_P considering $\lambda_P = 33 \text{ m}$.

different layered models, according to Fig. 11(a). The orange asterisks, the red dashed, the empty black dots and the magenta and green dashed curves correspond to models where the top and bottom layers (saturated sand) are characterized by $\sigma_f = 1.31, 3.6, 15.3, 110$ and 1000 mS m^{-1} , respectively, and the middle layer is a sandstone layer with $\sigma_f = 1.31 \text{ mS m}^{-1}$. The blue asterisks refer to the model with a clay-like material (Table 4) as intermediate layer and materials with $\sigma_f = 15.3 \text{ mS m}^{-1}$ in the top and bottom layers. Note that, even when σ_f was varied, we kept $\zeta = 11.51 \text{ mV}$ constant for sand layers, which is the value compatible with $\sigma_f = 1.31 \text{ mS m}^{-1}$ in our simulations (see Table B1). In the bottom part of Figs 12(a) and (b), specific regions along the vertical axis are highlighted to favour the distinction of lower amplitude values.

As expected, Fig. 12(a) shows that seismic waves are insensitive to σ_f . The amplitudes of u_z only change when a clay-like material is considered, because of its different mechanical properties compared with the sandstone (Table 4). On the other side, the amplitudes of V in Fig. 12(b) decrease with increasing $\sigma_{f\text{-sand}}$, which partially agrees with our experimental results. The increase in IR amplitudes observed experimentally when $\sigma_{f\text{-sand}}$ varied from 1.31 to 3.5 mS m^{-1} in Section 5 does not appear in the numerical data, using a sandstone layer, of Fig. 12(b). This is a consequence of using the same ζ for all sandstone simulations and, therefore, imposing that IR amplitudes only vary with $\sigma_{f\text{-sand}}$. Fig. 12(b) also shows that the higher the $\sigma_{f\text{-sand}}$

value (top and bottom layers), the lower the thickness leading to a maximum amplitude of the IR signals. A comparison between the u_z and V amplitudes when $d/\lambda_P = 1$ and $d/\lambda_P = 0.1$ in Fig. 12(b) shows that SE measurements become increasingly advantageous over conventional seismic reflection data for characterizing thin layers as $\sigma_{f\text{-sand}}$ increases, that is, with respect to the amplitude when $d/\lambda_P = 1$, the magnitude of V increases with $\sigma_{f\text{-sand}}$ when $d/\lambda_P = 0.1$ while u_z signals do not change.

Regarding the V amplitudes in Fig. 12(b), calculated by varying the $\sigma_{f\text{-sand}}$ values, we observe that the SE signals are sensitive to low σ_f layers at depths of about five times the P wavelength (here $\lambda_P \simeq 33 \text{ m}$), when the shallowest layer has σ_f values as high as 1 S m^{-1} . Thus, a robust measuring system, with a high dynamic range, should be able to pick up the lowest IR amplitudes predicted numerically (in the mV range) in field acquisitions. Moreover, as observed in our experiments in Section 4, for d values smaller than λ_P , the IR amplitudes are increased with respect to a reference IR created by a single layer, but the enhanced amplitudes will always be on the same order of magnitude as the reference IR. When d becomes very small the amplitudes decrease towards 0, as described by eq. (5). Notably, when $\sigma_f = 1.31 \text{ mS m}^{-1}$, the pattern of amplitudes in Fig. 12(b) closely resembles the observations made in Fig. 6, with a maximum amplitude increase also occurring when $d \simeq \lambda_P/2$. Finally, the amplification due to a clay layer shows how IRs can be used to detect geological structures associated with aquitards, as mentioned in

S.R. Pride & Garambois (2005), S.S. Haines & S.R. Pride (2006) and J.C. Dupuis *et al.* (2007).

7 CONCLUSION

We have studied the effect of layer thickness and fluid conductivity variation across interfaces on the EM interface responses with the aid of experimental and numerical data. The influence of these parameters on coseismic wavefields was also analysed. In our experiments and numerical simulations, we have followed the methodology introduced in V. Martins-Gomes *et al.* (2023) and used similar sand and sandstone materials. The experimental data obtained are unique since they only consider interfaces between two poroelastic media to study how the IR varies with d and contrasts of σ_f .

While exploring the influence of the thickness of a layer embedded in an otherwise homogeneous medium, we found that the maximum IR signal enhancement occurs when the layer thickness is about half the wavelength of the P -wave propagating through the layer. Both numerical and experimental data agree quite well, confirming that our observations are consistent with the theory developed by S. Pride (1994). In this study we were able to detect the experimental IR created by thicknesses equal to 0.2 times the wavelength of the P wave inside the thin layer. Our numerical study suggests that even for thicknesses around 0.1 times the wavelength of the P wave, the IR amplitudes would still be strong enough to be detected and in some cases (depending on the pore fluid conductivity contrast) the SE method could outperform the seismic response, which highlights the potential of IR signals to image very thin layers.

When studying the influence of fluid conductivity, the ability of Pride's theory to predict experimental data was reaffirmed by the satisfactory correlation between numerical and experimental records in terms of waveform and amplitude behaviour. This was achieved through simultaneous measurements of coseismic and IR signals, which allowed us to provide an original validation of Pride's theory using numerical simulations. Our unique experimental data complements previous studies on how the IR varies with σ_f . It should be noted, as mentioned by V. Martins-Gomes *et al.* (2023), that a more detailed quantitative analysis on how well theory can predict experiments will only be possible by precisely constraining the physical parameters of the medium studied. In addition, a thorough understanding of how contrasts of fluid conductivity influence IR signals is essential to facilitate the application of the SE method to ground water monitoring and geothermal reservoir investigation.

Using field-like numerical simulations we have compared our experimental results with a large-scale and low-frequency experiment that is similar to a real SE field acquisition. With that we have confirmed that the behaviour of the IR in our experiments will also be true to field data. Our numerical modelling predicts that even when the near surface layers are saturated with very conductive fluids ($\simeq 1 \text{ S m}^{-1}$), low σ_f layers at depths equivalent to five times the P wavelength could be detected by electric potential measurements using the three-electrode array discussed in M. Dietrich *et al.* (2018), whereas seismic waves are practically insensitive to σ_f contrasts. This highlights how the SE method, and more specifically the IR signals, may be successfully applied to near-surface investigations.

DATA AVAILABILITY

Both the data and the software used in this work can be made available upon reasonable request.

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REFERENCES

- Archie, G., 1942. The electrical resistivity log as an aid in determining some reservoir characteristics, *Trans. AIME*, **146**(01), 54–62.
- Barrière, J., Bordes, C., Brito, D. & Sénéchal, P., 2012. Laboratory monitoring of P waves in partially saturated sand, *Geophys. J. Int.*, **191**, 1152–1170.
- Bernardo, N., Martins-Gomes, V., Bordes, C. & Brito, D., 2024. Seismoelectric conversion at poroelastic/elastic interfaces and the role of dielectric permittivity: experimental and numerical analysis, *Geophys. Res. Lett.*, **51**(24), e2024GL112994.
- Biot, M.A., 1956a. Theory of propagation of elastic waves in a fluid-saturated porous solid. I. Low-frequency range, *J. acoust. Soc. Am.*, **28**(2), 168–178.
- Biot, M.A., 1956b. Theory of propagation of elastic waves in a fluid-saturated porous solid. II. Higher frequency range, *J. acoust. Soc. Am.*, **28**(2), 179–191.
- Biot, M.A., 1962. Mechanics of deformation and acoustic propagation in porous media, *J. Appl. Phys.*, **33**(4), 1482–1498.
- Block, G.I. & Harris, J.G., 2006. Conductivity dependence of seismoelectric wave phenomena in fluid-saturated sediments, *J. geophys. Res.*, **111**(B1), B01304.
- Bockris, J.O. & Reddy, A.K.N., 1970. The electrified interface, in *Modern Electrochemistry: An Introduction to an Interdisciplinary Area*, vol. 2, pp. 623–843, Springer.
- Bordes, C., Jouniaux, L., Dietrich, M., Pozzi, J.P. & Garambois, S., 2006. First laboratory measurements of seismo-magnetic conversions in fluid-filled Fontainebleau sand, *Geophys. Res. Lett.*, **33**(1), L01302.
- Bordes, C., Jouniaux, L., Garambois, S., Dietrich, M., Pozzi, J.P. & Gaffet, S., 2008. Evidence of the theoretically predicted seismo-magnetic conversion, *Geophys. J. Int.*, **174**(2), 489–504.
- Bordes, C., Sénéchal, P., Barrière, J., Brito, D., Normandin, E. & Jougnot, D., 2015. Impact of water saturation on seismoelectric transfer functions: a laboratory study of coseismic phenomenon, *Geophys. J. Int.*, **200**(3), 1317–1335.
- Bradford, J.H. & Deeds, J.C., 2006. Ground-penetrating radar theory and application of thin-bed offset-dependent reflectivity, *Geophysics*, **71**(3), K47–K57.
- Bucher, F., Monachesi, L.B., Castromán, G.A. & Zyserman, F.I., 2023. Seismoelectric response of 2-D elastic/poroelastic coupled media: a phenomenological approach, *Geophys. J. Int.*, **236**(1), 62–77.
- Butler, K.E., Russell, R.D., Kepic, A.W. & Maxwell, M., 1996. Measurement of the seismoelectric response from a shallow boundary, *Geophysics*, **61**(6), 1769–1778.
- Butler, K.E., Kullessa, B. & Pugin, A.J.M., 2018. Multimode seismoelectric phenomena generated using explosive and vibroseis sources, *Geophys. J. Int.*, **213**(2), 836–850.

- Carabelli, E. & Folicaldi, R., 1957. Seismic model experiment on thin layers, *Geophys. Prospect.*, **5**(3), 317–327.
- Chen, B. & Mu, Y., 2005. Experimental studies of seismoelectric effects in fluid-saturated porous media, *J. Geophys. Eng.*, **2**(3), 222–230.
- de Voogd, N. & den Rooijen, H., 1983. Thin-layer response and spectral bandwidth, *Geophysics*, **48**(1), 12–18.
- Deparis, J. & Garambois, S., 2009. On the use of dispersive APVO GPR curves for thin-bed properties estimation: theory and application to fracture characterization, *Geophysics*, **74**(1), J1–J12.
- Devi, M.S., Garambois, S., Brito, D., Dietrich, M., Poydenot, V. & Bordes, C., 2018. A novel approach for seismoelectric measurements using multielectrode arrangements: II—Laboratory measurements, *Geophys. J. Int.*, **214**(3), 1783–1799.
- Dietrich, M., Devi, M.S., Garambois, S., Brito, D. & Bordes, C., 2018. A novel approach for seismoelectric measurements using multielectrode arrangements - I: Theory and numerical experiments, *Geophys. J. Int.*, **215**(1), 61–80.
- Domra Kana, J., Djongyang, N., Raïdandi, D., Njandjock Nouck, P. & Dadjé, A., 2015. A review of geophysical methods for geothermal exploration, *Renew. Sustain. Energy Rev.*, **44**, 87–95.
- Dukhin, A., Dukhin, S. & Goetz, P., 2005. Electrokinetics at high ionic strength and hypothesis of the double layer with zero surface charge, *Langmuir*, **21**(22), 9990–9997.
- Dupuis, J.C., Butler, K.E. & Kepic, A.W., 2007. Seismoelectric imaging of the vadose zone of a sand aquifer, *Geophysics*, **72**(6), A81–A85.
- Dupuis, J.C., Butler, K.E., Kepic, A.W. & Harris, B.D., 2009. Anatomy of a seismoelectric conversion: Measurements and conceptual modeling in boreholes penetrating a sandy aquifer, *J. geophys. Res.*, **114**(B10), B10306.
- Dzieran, L., Thorwart, M., Rabbel, W. & Ritter, O., 2019. Quantifying interface responses with seismoelectric spectral ratios, *Geophys. J. Int.*, **217**(1), 108–121.
- Garambois, S. & Dietrich, M., 2001. Seismoelectric wave conversions in porous media: Field measurements and transfer function analysis, *Geophysics*, **66**(5), 1417–1430.
- Garambois, S. & Dietrich, M., 2002. Full waveform numerical simulations of seismoelectromagnetic wave conversions in fluid-saturated stratified porous media, *J. geophys. Res.: Solid Earth*, **107**(B7), ESE 5–1–ESE 5–18.
- Garambois, S., Sénéchal, P. & Perroud, H., 2002. On the use of combined geophysical methods to assess water content and water conductivity of near-surface formations, *J. Hydrol.*, **259**(1–4), 32–48.
- Goldman, M. & Neubauer, F., 1994. Groundwater exploration using integrated geophysical techniques, *Surv. Geophys.*, **15**, 331–361.
- Grobbe, N. & Slob, E.C., 2016. Seismo-electromagnetic thin-bed responses: natural signal enhancements?, *J. geophys. Res.: Solid Earth*, **121**(4), 2460–2479.
- Grobbe, N., Slob, E.C. & Thorbecke, J.W., 2016. Comparison of eigenvectors for coupled seismo-electromagnetic layered-Earth modelling, *Geophys. J. Int.*, **206**(1), 152–190.
- Guichet, X., Jouniaux, L. & Pozzi, J.P., 2003. Streaming potential of a sand column in partial saturation conditions, *J. geophys. Res.: Solid Earth*, **108**(B3), e2001JB001517.
- Guichet, X., Jouniaux, L. & Catel, N., 2006. Modification of streaming potential by precipitation of calcite in a sandwater system: laboratory measurements in the pH range from 4 to 12, *Geophys. J. Int.*, **166**(1), 445–460.
- Guo, J., Li, Y., Wang, Y., Lyu, Q. & Zhou, L., 2023. Identification of an oil-bearing layer by formation water resistivity: a case study of the jurassic reservoir, southwest Ordos basin, *Geophysics*, **88**(4), B151–B162.
- Haartsen, M.W. & Pride, S.R., 1997. Electro seismic waves from point sources in layered media, *J. geophys. Res.: Solid Earth*, **102**(B11), 24 745–24 769.
- Haines, S.S. & Pride, S.R., 2006. Seismoelectric numerical modeling on a grid, *Geophysics*, **71**(6).
- Han, L., Huang, X., Quintal, B., Ji, Y. & Sun, T., 2024. Seismoelectric wave propagation in velocity and attenuation anisotropic media, *IEEE Trans. Geosci. Remote Sens.*, **62**, 1–10.
- Holzhauser, J., 2015. *Seismic and seismoelectric monitoring of an consolidated porous medium*, Ph.D. thesis, Université de Pau et des Pays de l'Adour.
- Holzhauser, J., Brito, D., Bordes, C., Brun, Y. & Guatarbes, B., 2017. Experimental quantification of the seismoelectric transfer function and its dependence on conductivity and saturation in loose sand, *Geophys. Prospect.*, **65**(4), 1097–1120.
- Hu, K., Ren, H., Huang, Q., Zeng, L., Butler, K.E., Jougnot, D., Linde, N. & Holliger, K., 2023. Water table and permeability estimation from multi-channel seismoelectric spectral ratios, *J. geophys. Res.: Solid Earth*, **128**(5), e2022JB025505.
- Ishido, T. & Mizutani, H., 1981. Experimental and theoretical basis of electrokinetic phenomena in rock-water systems and its applications to geophysics, *J. geophys. Res.: Solid Earth*, **86**(B3), 1763–1775.
- Jardani, A. & Revil, A., 2015. Seismoelectric couplings in a poroelastic material containing two immiscible fluid phases, *Geophys. J. Int.*, **202**(2), 850–870.
- Jougnot, D. & Solazzi, S.G., 2021. Predicting the frequency-dependent effective excess charge density: a new upscaling approach for seismoelectric modeling, *Geophysics*, **86**(5), WB19–WB28.
- Jougnot, D., Rubino, J., Rosas-Carbajal, M., Linde, N. & Holliger, K., 2013. Seismoelectric effects due to mesoscopic heterogeneities, *Geophys. Res. Lett.*, **40**, 2033–2037.
- Kang, Y., Wei, P. & Li, Y., 2024. Seismoelectric wave propagation through a fluid-saturated porous sandwiched interlayer, *Appl. Math. Modell.*, **131**, 469–486.
- Li, J. et al., 2024. Time-domain finite element method based on arbitrary quadrilateral meshes for two-dimensional SHTE mode seismoelectric and electroseismic waves modelling, *Geophys. Prospect.*, **72**(6), 2419–2438.
- Li, S., Collini, H. & Jackson, M.D., 2018. Anomalous Zeta potential trends in natural sandstones, *Geophys. Res. Lett.*, **45**(20), 11 068–11 073.
- Liu, J., Gao, Y., Wang, J., Zhu, T., Ding, P., Zhou, G. & Li, S., 2024. Seismoelectric imaging of aquifer interface: theoretical investigation and laboratory experiments, *Geophys. Res. Lett.*, **51**(23), e2024GL112534.
- Liu, Y. & Schmitt, D.R., 2003. Amplitude and AVO responses of a single thin bed, *Geophysics*, **68**(4), 1161–1168.
- Lorne, B., Perrier, F. & Avouac, J.P., 1999a. Streaming potential measurements: 2. Relationship between electrical and hydraulic flow patterns from rock samples during deformation, *J. geophys. Res.: Solid Earth*, **104**(B8), 17879–17896.
- Lorne, B., Perrier, F. & Avouac, J.P., 1999b. Streaming potential measurements: 1. Properties of the electrical double layer from crushed rock samples, *J. geophys. Res.: Solid Earth*, **104**(B8), 17 857–17 877.
- Ma, X., Liu, Y., Yin, C., Zhang, B. & Ren, X., 2022. Estimation of fluid salinity using coseismic electric signal generated by an earthquake, *Geophys. J. Int.*, **233**(1), 127–144.
- Martins-Gomes, V., 2022. *Experimental characterization and modelling of electromagnetic waves generated by seismoelectric conversion at porous media interfaces*, Ph.D. thesis, Université de Pau et des Pays de l'Adour.
- Martins-Gomes, V., Brito, D., Garambois, S., Dietrich, M., Bordes, C. & Barucq, H., 2023. Seismoelectric wave conversions at an interface: a quantitative comparison between laboratory data and full-waveform modelling, *Geophys. J. Int.*, **235**(3), 2992–3011.
- Martner, S.T. & Sparks, N.R., 1959. The electroseismic effect, *Geophysics*, **24**(2), 297–308.
- Monachesi, L.B., Zyserman, F.I., Jouniaux, L. & Thompson, A.H., 2023. Electromagnetic/acoustic coupling in partially saturated porous rocks: an extension of Pride's theory, *Transport Porous Media*, **149**(3), 785–815.
- Morency, C. et al., 2021. Seismoelectric effects for geothermal resources assessment and monitoring, in *SEG/AAPG/SEPM First International Meeting for Applied Geoscience & Energy*, Society of Exploration Geophysicists, Denver, United States.

- Nie, C., Wang, J., Guan, W. & Hu, H., 2025. Experimental study of seismoelectric response in partially saturated rock: a fluid saturated/unsaturated interface model, *IEEE Trans. Geosci. Remote Sens.*, **63**, 1–8.
- Nouradine, H., Schamper, C., Valdes, D., Moussa, I., Ramel, D. & Plagnes, V., 2024. Integrating geological, hydrogeological and geophysical data to identify groundwater resources in granitic basement areas (Guéra Massif, Chad), *Hydrogeol. J.*, **32**(3), 759–784.
- Papageorgiou, G. & Chapman, M., 2021. Seismic tuning of dispersive thin layers, *Geophys. Prospect.*, **69**(3), 622–628.
- Peng, R., Di, B., Wei, J., Ding, P., Zhao, J., Pan, X. & Liu, Z., 2017. Experimental study of the seismoelectric interface response in wedge and cavity models, *Geophys. J. Int.*, **210**(3), 1703–1720.
- Pride, S., 1994. Governing equations for the coupled electromagnetics and acoustics of porous media, *Phys. Rev. B*, **50**(21), 15 678–15 696.
- Pride, S.R., 2005. Relationships between seismic and hydrological properties, in *Hydrogeophysics*, pp. 253–290, eds Rubin, Y. & Hubbard, S.S., Springer.
- Pride, S.R. & Garambois, S., 2005. Electrostatic wave theory of Frenkel and more recent developments, *J. Eng. Mech.*, **131**(9), 898–907.
- Pride, S.R. & Haartsen, M.W., 1996. Electrostatic wave properties, *J. acoust. Soc. Am.*, **100**(3), 1301–1315.
- Pride, S.R. & Morgan, F.D., 1991. Electrokinetic dissipation induced by seismic waves, *Geophysics*, **56**(7), 914–925.
- Pride, S.R., Gangi, A.F. & Morgan, F.D., 1992. Deriving the equations of motion for porous isotropic media, *J. acoust. Soc. Am.*, **92**(6), 3278–3290.
- Rabbal, W., Iwanowski-Strahser, K., Strahser, M., Dzieran, L. & Thorwart, M., 2020. Seismoelectric field measurements in unconsolidated sediments in comparison with other methods of near-surface prospecting, *Seismoelectric Exploration*, chap. 22, pp. 347–363, American Geophysical Union.
- Ren, H., Huang, Q. & Chen, X., 2010. A new numerical technique for simulating the coupled seismic and electromagnetic waves in layered porous media, *Earthq. Sci.*, **23**(2), 167–176.
- Ren, H., Huang, Q. & Chen, X., 2016. Existence of evanescent electromagnetic waves resulting from seismoelectric conversion at a solid-porous interface, *Geophys. Int.*, **204**(1), 147–166.
- Ren, H., Huang, Q. & Chen, X., 2018. Quantitative understanding on the amplitude decay characteristic of the evanescent electromagnetic waves generated by seismoelectric conversion, *Pure appl. Geophys.*, **175**(8), 2853–2879.
- Revil, A. & Florsch, N., 2010. Determination of permeability from spectral induced polarization in granular media, *Geophys. J. Int.*, **181**(3), 1480–1498.
- Revil, A. & Glover, P.W.J., 1998. Nature of surface electrical conductivity in natural sands, sandstones, and clays, *Geophys. Res. Lett.*, **25**(5), 691–694.
- Revil, A. & Mahardika, H., 2013. Coupled hydromechanical and electromagnetic disturbances in unsaturated porous materials, *Water Resour. Res.*, **49**(2), 744–766.
- Revil, A., Pezard, P.A. & Glover, P.W.J., 1999. Streaming potential in porous media: 1. Theory of the zeta potential, *J. geophys. Res.: Solid Earth*, **104**(B9), 20 021–20 031.
- Robinson, E.A., 1985. Seismic time-invariant convolutional model, *Geophysics*, **50**(12), 2742–2751.
- Rosas-Carbajal, M., Jougnot, D., Rubino, J.G., Monachesi, L., Linde, N. & Holliger, K., 2020. Seismoelectric Signals Produced by Mesoscopic Heterogeneities, in *Seismoelectric Exploration*, chap. 19, pp. 269–287, American Geophysical Union.
- Russell, R.D., Butler, K.E., Kepic, A.W. & Maxwell, M., 1997. Seismoelectric exploration, *The Leading Edge*, **16**(11), 1611–1615.
- Saunders, J.H., Jackson, M.D. & Pain, C.C., 2008. Fluid flow monitoring in oil fields using downhole measurements of electrokinetic potential, *Geophysics*, **73**(5), E165–E180.
- Schakel, M. & Smeulders, D., 2010. Seismoelectric reflection and transmission at a fluid/porous-medium interface, *J. acoust. Soc. Am.*, **127**(1), 13–21.
- Schakel, M.D., Smeulders, D.M.J., Slob, E.C. & Heller, H.K.J., 2011. Seismoelectric interface response: Experimental results and forward model, *Geophysics*, **76**(4), N29–N36.
- Sénéchal, P., Perroud, H., Kedziorek, M.A.M., Bourg, A.C.M. & Gloaguen, E., 2005. Non destructive geophysical monitoring of water content and fluid conductivity anomalies in the near surface at the border of an agricultural field, *Subsurf. Sens. Technol. Appl.*, **6**, 167–192.
- Shakas, A. & Linde, N., 2015. Effective modeling of ground penetrating radar in fractured media using analytic solutions for propagation, thin-bed interaction and dipolar scattering, *J. Appl. Geophys.*, **116**, 206–214.
- Shen, C., Brito, D., Diaz, J., Sanjuan, F., Bordes, C. & Garambois, S., 2022. Pulsed-laser source characterization in laboratory seismic experiments, *Geomech. Geophys. Geo-Energy Geo-Resour.*, **8**(1), 1–26.
- Slattery, J.C., 1967. Flow of viscoelastic fluids through porous media, *AIChE J.*, **13**(6), 1066–1071.
- Smeulders, D., Grobde, N., Heller, H. & Schakel, M., 2014. Seismoelectric conversion for the detection of porous medium interfaces between wetting and nonwetting fluids, *Vadose Zone J.*, **13**(5), doi: 10.2136/vzj2013.06.0106.
- Stern, O., 1924. Zur theorie der elektrolytischen doppelschicht, *Z. Elektrochem. angew. phys. Chem.*, **30**(21–22), 508–516.
- Sun, Y.C. et al., 2019. Numerical simulations to explain the coseismic electromagnetic signals: a case study for a M5.4 aftershock of the 2016 Kumamoto earthquake, *Earth Planets Space*, **71**(1), doi: 10.1186/s40623-019-1122-7.
- Tatalovic, R., McDonald, J. & Gardner, G., 1988. Modelling study of tuning effects for the calculation of the thickness and areal extent of thin beds, *First Break*, **6**(12), 395–404.
- Thompson, R.R., 1936. The seismic electric effect, *Geophysics*, **1**(3), 327–335.
- Vinogradov, J., Jaafar, M.Z. & Jackson, M.D., 2010. Measurement of streaming potential coupling coefficient in sandstones saturated with natural and artificial brines at high salinity, *J. geophys. Res.: Solid Earth*, **115**(B12), e2010JB007593.
- Vinogradov, J., Jackson, M.D. & Chamerois, M., 2018. Zeta potential in sandpacks: effect of temperature, electrolyte pH, ionic strength and divalent cations, *Coll. Surf. A: Physicochem. Eng. Aspects*, **553**, 259–271.
- Walker, E. & Glover, P., 2018. Measurements of the relationship between microstructure, pH, and the streaming and zeta potentials of sandstones, *Transport Porous Media*, **121**(1), 183–206.
- Walton, K., 1987. The effective elastic moduli of a random packing of spheres, *J. Mech. Phys. Solids*, **35**(2), 213–226.
- Wang, D., Gao, Y., Tong, P., Wang, J., Yao, C. & Wang, B., 2021. Electrostatic and seismoelectric responses at irregular interfaces: possible application to reservoir exploration, *J. Petrol. Sci. Eng.*, **202**, 108513.
- Wang, D., Gao, Y., Zhou, G., Tong, P., Cheng, Q., Yao, C. & Bai, X., 2023. Finite-element modelling of seismoelectric and electrostatic waves in frequency domain: 2-D SHTE mode, *Geophys. J. Int.*, **234**(3), 2306–2327.
- Wang, J., Zhu, Z., Gao, Y., Morgan, F.D. & Hu, H., 2020. Measurements of the seismoelectric responses in a synthetic porous rock, *Geophys. J. Int.*, **222**(1), 436–448.
- Whitaker, S., 1969. Advances in theory of fluid motion in porous media, *Industr. Eng. Chem.*, **61**(12), 14–28.
- Widess, M.B., 1973. How thin is a thin bed?, *Geophysics*, **38**(6), 1176–1180.
- Yuan, S., Ren, H., Huang, Q., Zheng, X.Z., Yang, Z., He, Z., Zhang, W. & Chen, X., 2021. Refining higher modes of Rayleigh waves using seismoelectric signals excited by a weight-drop source: study from numerical simulation aspect, *J. geophys. Res.: Solid Earth*, **126**(5), e2020JB021336.
- Zhao, Y., Sun, X. & Nie, Z., 2023. Study of the seismoelectric effect in saturated porous media using a bundle of capillary tubes model, *Electronics*, **12**(2), doi: 10.3390/electronics12020379.

Zheng, X.Z., Ren, H., Butler, K.E., Zhang, H., Sun, Y.C., Zhang, W., Huang, Q. & Chen, X., 2021. Seismoelectric and electroseismic modeling in stratified porous media with a shallow or ground surface source, *J. geophys. Res.: Solid Earth*, **126**(9), e2021JB021950.

Zhu, Z. & Nafi Toksöz, M., 2013. Experimental measurements of the streaming potential and seismoelectric conversion in Berea sandstone, *Geophys. Prospect.*, **61**(3), 688–700.

Zhu, Z. & Toksöz, M.N., 2003. *Effects of Saturant Conductivity on Seismoelectric Conversion*, Tech. Rep., Massachusetts Institute of Technology. Available at: <https://dspace.mit.edu/handle/1721.1/67874>

Zhu, Z., Haartsen, M.W. & Toksöz, M.N., 1999. Experimental studies of electrokinetic conversions in fluid-saturated borehole models, *Geophysics*, **64**(5), 1349–1356.

Zhu, Z., Haartsen, M.W. & Toksöz, M.N., 2000. Experimental studies of seismoelectric conversions in fluid-saturated porous media, *J. geophys. Res.: Solid Earth*, **105**(B12), 28 055–28 064.

Zyserman, F.I., Gauzellino, P.M. & Santos, J.E., 2010. Finite element modeling of SHTE and PSVTM electroseismics, *J. Appl. Geophys.*, **72**(2), 79–91.

APPENDIX A: PHYSICAL PARAMETERS OF THE ELECTROKINETIC TF FOR COMPRESSIONAL WAVES

The variables controlling the transfer function ratios $\mathbf{E}/\mathbf{u}$, $\mathbf{E}/\dot{\mathbf{u}}$ and $\mathbf{E}/\ddot{\mathbf{u}}$ in eq. (8) are further detailed in this appendix. $\tilde{\rho}$ is the effective complex fluid density which controls the dynamic fluid flow relative to the solid grains ($\mathbf{w}$):

$$\tilde{\rho}(\omega) = \frac{i}{\omega} \frac{\eta}{k(\omega)}. \quad (\text{A1})$$

From its definition above, $\tilde{\rho}$ not only depends on the fluid viscosity but also on the dynamic permeability $k(\omega)$. The latter is defined by S. Pride (1994) as:

$$k(\omega) = k_0 \left[\left(1 - i \frac{4}{m} \frac{\omega}{\omega_t} \right)^{1/2} - i \frac{\omega}{\omega_t} \right]^{-1}, \quad (\text{A2})$$

with m and k_0 specified in table A1, and ω_t is the angular frequency that marks the transition from viscous (low-frequency) to inertial (high-frequency) flow:

$$\omega_t = \frac{\phi}{\alpha_\infty k_0} \frac{\eta}{\rho_f}. \quad (\text{A3})$$

Additionally, $f_t = \omega_t/2\pi$ is the transition frequency in Hertz.

The effective complex electrical permittivity of the porous media $\tilde{\epsilon}$, that controls the degree with which the medium is polarized due to an applied electrical field, is expressed by S.R. Pride & M.W. Haartsen (1996) as:

$$\tilde{\epsilon}(\omega) = \epsilon + \frac{i}{\omega} \sigma(\omega) - \tilde{\rho}(\omega) L^2(\omega), \quad (\text{A4})$$

and appears naturally when manipulating Pride's equations. The porous medium electrical permittivity ϵ is usually also complex and frequency-dependent, however the governing equations in S. Pride (1994) consider only media in which the electric-double layer is much thinner than the radii of rock grains, and those where $\kappa_s/\kappa_f \ll 1$ (i.e. with a large dielectric constant contrast between solid and fluid phases). When these conditions are met in a fully saturated medium, ϵ is calculated with:

$$\epsilon = \epsilon_0 \left[\frac{\phi}{\alpha_\infty} (\kappa_f - \kappa_s) + \kappa_s \right]. \quad (\text{A5})$$

The total conductivity of the porous medium, $\sigma(\omega)$ in eq. (A4), considering surface-conduction effects and with a

weak-frequency dependence is:

$$\sigma(\omega) = \frac{\phi \sigma_f}{\alpha_\infty} \left[1 + \frac{2}{\sigma_f \Lambda} [C_{\text{em}} + C_{\text{os}}(\omega)] \right], \quad (\text{A6})$$

where C_{em} and C_{os} contribute to the surface conductance by accounting, respectively, for electromigration and electroosmosis of ions comprised in the double layer. Their definition is given by eqs (195) and (206) in S. Pride (1994). The other unknown in eq. (A6) is Λ whose definition is analogous to the pore radius and is usually obtained from the relation: $k_0 = (1/m)(\phi/\alpha_\infty)\Lambda^2$.

All coupling between Biot's and Maxwell's equations is controlled by the coefficient $L(\omega)$:

$$L(\omega) = L_0 \left[1 - i \frac{\omega}{\omega_t} \frac{m}{4} \left(1 - 2 \frac{\tilde{d}}{\Lambda} \right)^2 \left(1 - i^{3/2} \tilde{d} \sqrt{\frac{\omega \rho_f}{\eta}} \right)^2 \right]^{-1/2} \quad (\text{A7})$$

where $\tilde{d}$ refers to the Debye length, described by eqs (29) and (210) of S. Pride (1994), and defined as a measure of the diffuse double layer thickness. L_0 is the low-frequency (i.e. $f \ll f_t$) coupling coefficient defined as:

$$L_0 = -\frac{\phi}{\alpha_\infty} \frac{\epsilon_0 \kappa_f \zeta}{\eta} \left(1 - 2 \frac{\tilde{d}}{\Lambda} \right) \Leftrightarrow -\frac{\phi}{\alpha_\infty} C_{ek} \sigma_f \left(1 - 2 \frac{\tilde{d}}{\Lambda} \right), \quad (\text{A8})$$

with C_{ek} referring to the electrokinetic coupling, related to the streaming potential by admitting the validity of the Helmholtz-Smoluchowski model.

For consistency with the sand and sandstone media used in our experiments, we chose to work with a definition of C_{ek} , that accounts for surface conductivity (σ_n) (X. Guichet *et al.* 2003, 2006):

$$C_{ek} = \frac{\epsilon_0 \kappa_f \zeta}{\eta \sigma_f} \frac{1}{1 + m_c \left(\frac{\alpha_\infty}{\phi} - 1 \right) \frac{\sigma_n}{\sigma_f}}. \quad (\text{A9})$$

Regarding the cementation exponent m_c (see G. Archie 1942) in this equation, we define it here as $m_c = (\log(\phi) - \log(\alpha_\infty))/\log(\phi)$.

The last term to be discussed from the TF definition is $\beta_L(\omega)$, where the subscript L refers to longitudinal waves. S.R. Pride & M.W. Haartsen (1996) write it as:

$$\beta_L(\omega) = -\frac{H s_\xi^2 - \rho}{C s_\xi^2 - \rho_f}, \quad (\text{A10})$$

with $\xi = P$ or SP referring to the fast P -wave and the slow compressional (or Biot's) wave, respectively. In this equation $\rho = \phi \rho_f + (1 - \phi) \rho_s$ is the bulk density, s_ξ the phase slowness of a given wave type and H and C are poromechanic factors detailed further on. Physically, β_L relates the amount of relative fluid displacement ($\mathbf{w}$) with the solid grain displacement ($\mathbf{u}$) of a porous medium.

Before explaining s_ξ , five poromechanic coefficients [after M.A. Biot (1962) and S.R. Pride *et al.* (1992)], depending on fluid, solid and (drained) frame bulk moduli, as well as porosity, will be presented once they largely simplify the mechanical equations and consequently the system matrix $\mathbf{A}$ discussed in the next appendix. Following S. Pride (1994):

$$D = \frac{K_f}{\phi K_s^2} [(1 - \phi) K_s - K_{fr}], \quad (\text{A11a})$$

$$K_G = \frac{K_{fr} + \phi K_f + (1 + \phi) K_s D}{1 + D}, \quad (\text{A11b})$$

$$C = \frac{K_f + K_s D}{1 + D}, \quad (\text{A11c})$$

$$M = \frac{K_f}{\phi(1+D)}, \quad (\text{A11d})$$

$$H = K_G + \frac{4}{3}G_{fr}. \quad (\text{A11e})$$

Now the phase slowness of each longitudinal wave type can be introduced as derived in S.R. Pride & M.W. Haartsen (1996):

$$2s_P^2 = \gamma - \sqrt{\gamma^2 - \frac{4\bar{\rho}\rho}{\Delta} \left(\frac{\rho_t}{\rho} + \frac{\bar{\rho}L^2}{\bar{\epsilon}} \right)}, \quad (\text{A12a})$$

$$2s_{PS}^2 = \gamma + \sqrt{\gamma^2 - \frac{4\bar{\rho}\rho}{\Delta} \left(\frac{\rho_t}{\rho} + \frac{\bar{\rho}L^2}{\bar{\epsilon}} \right)}. \quad (\text{A12b})$$

In the equations above $\rho_t = \rho - \rho_f^2/\bar{\rho}$ is the complex density, and the definition $\Delta = HM - C^2$ is used for convenience as well as the auxiliary parameter:

$$\gamma = \frac{\rho M + \bar{\rho}H(1 + \bar{\rho}L^2/\bar{\epsilon}) - 2\rho_f C}{\Delta}. \quad (\text{A13})$$

From s_ξ , it is straightforward to calculate the real phase velocity and intrinsic attenuation α_ξ (M.A. Biot 1956a, b):

$$v_\xi = 1/\text{Re}\{s_\xi\}, \quad (\text{A14a})$$

$$\alpha_\xi = \omega \text{Im}\{s_\xi\}. \quad (\text{A14b})$$

For completeness and to help calculate the intrinsic attenuation of EM-waves generated at interfaces, the equation that gives the slowness of EM waves in saturated porous media is:

$$2s_{EM}^2 = \frac{\rho_t}{G_{fr}} + \mu\bar{\epsilon} \left(1 + \frac{\bar{\rho}L^2}{\bar{\epsilon}} \right) - \sqrt{\left[\frac{\rho_t}{G_{fr}} \mu\bar{\epsilon} \left(1 + \frac{\bar{\rho}L^2}{\bar{\epsilon}} \right) \right]^2 - 4\mu \frac{\rho_f^2 L^2}{G_{fr}}}. \quad (\text{A15})$$

Finally, v_{EM} and α_{EM} can be calculated, respectively, with eqs (A14a) and (A14b) above.

APPENDIX B: TABLE OF PHYSICAL PARAMETERS OF THE SAND AND SANDSTONE MATERIALS USED IN THE NUMERICAL SIMULATIONS

Below we reproduce the table of V. Martins-Gomes *et al.* (2023) that contains the physical parameters used here for the sand and sandstone materials, unless told otherwise. See Section 3 for further details about the physical parameters used in the numerical modelling. Because we have used here the same media as V. Martins-Gomes *et al.* (2023), we refer to that work for more details about the actual media used.

Table B1. Physical parameters adopted for the sand and sandstone materials used in the SE experiments and numerical simulations. Relevant constants are also included in the list. * refers to dimensionless properties. † refers to standard values.

Name, symbol, units	Sand	Sandstone	References	Sandstone
Porosity (ϕ)[*]	0.4	0.24	J. Barrière <i>et al.</i> (2012)	Measured
Saturation (S_w)[*]	1		–	
DC permeability (k_0)(m ²)	1×10^{-11}	7×10^{-15}	J. Barrière <i>et al.</i> (2012)	B. Lorne <i>et al.</i> (1999a)
Bulk (solid) modulus (K_s)(Pa)	3.6×10^{10}	3.6×10^{10}	†	
Shear (solid) modulus (G_s)(Pa)	4.4×10^{10}	4.4×10^{10}	†	
Bulk (fluid) modulus (K_f)(Pa)	2.5×10^9	2.5×10^9	†	
Bulk (frame) modulus (K_{fr})(Pa)	2.38×10^7	3.41×10^8	S.R. Pride (2005)	Seismic velocities
Shear (frame) modulus (G_{fr})(Pa)	1.43×10^7	2.10×10^8	S.R. Pride (2005)	
P-wave velocity (v_P)(m s ⁻¹)	1780	2200	From the first breaks	
EM-wave velocity (v_{EM})(m s ⁻¹)	5.4×10^7	8.5×10^7	S.R. Pride & M.W. Haartsen (1996)	
S-wave velocity (v_S)(m s ⁻¹)	90.4	336.8	S.R. Pride & M.W. Haartsen (1996)	
Fluid viscosity (η)(Pa s)		0.001	†	
Solid density (ρ_s)(kg m ⁻³)	2650	2120	†	Measured
Fluid density (ρ_f)(kg m ⁻³)		1000	†	
Fluid Conductivity (σ_f)(mS m ⁻¹)		1.2–1.8	Measured	
Molarity (C_0)(mmol l ⁻¹)		0.128–0.195	S. Pride (1994): $C_0 = \sigma_f/(e^2(b_{Na} + b_{Cl})N_A)$	
Surface Conductivity (σ_n)(mS m ⁻¹)	0.083	2	A. Revil & P.W.J. Glover (1998)	B. Lorne <i>et al.</i> (1999a)
Temperature (T)(K)		288.35	Measured	
Fluid permittivity (κ_f)[*]		80	†	
Solid permittivity (κ_s)[*]		4	†	
Tortuosity (α_∞)[*]	1.65	3	J. Holzhauser (2015)	B. Lorne <i>et al.</i> (1999a)
Pore geometry number (m)[*]		6	S. Pride (1994)	M.W. Haartsen & S.R. Pride (1997)
Zeta potential (ζ)(mV)	6.79–12.91	-12	X. Guichet <i>et al.</i> (2003): $\zeta = -14.6 \ln(\sigma_f) - 85.4$	B. Lorne <i>et al.</i> (1999a)
Transition frequency (f_t)(kHz)	3.86	1822.56	$\omega_t/2\pi$, equation (A3)	
Constants				
Avogadro's number (N_A)(mol ⁻¹)		6.02×10^{23}	–	
Elementary charge (e)(C)		1.60×10^{-19}	–	
Ionic mobility ($b_{I=Na=Cl}$)(m (sN) ⁻¹)		3×10^{11}	Approximation, after S. Pride (1994), for typical inorganic ion	
Vacuum permittivity (ϵ_0)(F m ⁻¹)		8.85×10^{-12}	–	
Vacuum permeability (μ_0)(H m ⁻¹)		$4\pi \times 10^{-7}$	–	
Boltzmann constant (k_b)(J K ⁻¹)		1.38×10^{-23}	–	

Table C1. Values of the consolidation factor (c) used to calculate the frame moduli of each sandstone layer in Section 4 through expressions (C1). Sandstone are shown with respect to their thickness d and * indicates that c is dimensionless.

Sandstone thickness(d)(cm)	0.2	0.5	0.74	0.9	1.63	1.91	2.5	5
Consolidation factor(c)[*]	330	330	330	330	330	50.495	330	58.075

Table C2. Effective pressure (p_{eff}) used to calculate the frame moduli of the sand layers in the three-layer models of Section 4 with expressions (C2). Each model is referenced by the thickness of its sandstone layer and we follow the labelling of Table 1 for the sand layers.

Reference sandstone thickness (cm)	0.2	0.5	0.74	0.9	1.63	1.91	2.5	5
p_{eff} - I1 (kPa)	2.453	363.302	2.453	2.453	48.287	259.063	2.453	2.453
p_{eff} - I3 (kPa)	2.453	363.302	2.453	2.453	48.287	93.508	2.453	2.453

APPENDIX C: ESTIMATING THE FRAME MODULI OF THE EXPERIMENTAL MEDIA

In order to estimate the frame moduli K_{fr} and G_{fr} of the sand and sandstone materials used in our experiments, we have used the models of S.R. Pride (2005) for unconsolidated and consolidated media and the P -wave velocities obtained from our experimental data. The porosity and the bulk and shear moduli of the porous matrix are those shown in Table B1.

For consolidated media, the following expressions have been used:

$$K_{\text{fr}} = K_s \frac{1 - \phi}{1 + c\phi} \quad \text{and} \quad G_{\text{fr}} = G_s \frac{1 - \phi}{1 + 1.5c\phi}, \quad (\text{C1})$$

where c is a consolidation factor which is inversely proportional to the degree of consolidation of the medium.

When calculating the P -wave velocity using eqs (A12a) and (A14a), the value of c used in the expressions (C1) was chosen to ensure that the v_p predicted by Pride's theory is equal to the experimental velocities in the frequency range of the experimental data. For the sandstone layers in Table 1 (I2) the frame moduli was obtained with the c shown in Table C1.

For unconsolidated media, the model presented in S.R. Pride (2005) was built upon the developments of K. Walton (1987) in which the media are described as randomly distributed spheres. Inside such media, the number of grain-to-grain contacts n_o increases with the effective pressure p_{eff} . The latter is the difference between the confining (p_c) and the fluid (p_f) pressure, that

is, $p_{\text{eff}} = p_c - p_f$. Because K_{fr} is defined under a drained condition we consider $p_{\text{eff}} = p_c$. Using these ideas, the frame moduli expressions are:

$$K_{\text{fr}} = \frac{1}{6} \left[\frac{4(1 - \phi)^2 n_o^2 p_o}{\pi^4 C_s^2} \right]^{1/3} \frac{(p_{\text{eff}}/p_o)^{1/2}}{\left[1 + (16 \frac{p_{\text{eff}}}{9p_o})^4 \right]^{1/24}} \quad \text{and} \quad G_{\text{fr}} = 3 \frac{K_{\text{fr}}}{5}. \quad (\text{C2})$$

In these expression, the limiting pressure p_o defines the value beyond which no new grain contacts can take place. The parameter C_s is a compliance factor, inversely proportional to the solid grain bulk and shear moduli, defined by: $C_s = \frac{1}{4\pi} \left(\frac{1}{G_s} + \frac{1}{K_s + G_s/3} \right)$.

It follows that the free parameters of the unconsolidated expression above are n_o , p_o and p_{eff} . However, we have chosen to follow the suggestion of S.R. Pride (2005) for sand media and fix $n_o = 9$ and $p_o = 18$ MPa. Therefore p_{eff} becomes the only free parameter.

To calculate the frame moduli of the sand layers using expressions (C2), for the three-layer models of Table 1 (I1 and I3), we have used the p_{eff} values in Table C2. Using these values, in the frequency range of our measurements the P -wave velocities v_p obtained with Pride's theory agree with our experimental velocities.